

Bank Liquidity Creation

Allen N. Berger[†] and Christa H.S. Bouwman[‡]

January 2007

Although the modern theory of financial intermediation portrays liquidity creation as an essential service provided by banks, comprehensive measures of bank liquidity creation do not exist. We therefore have little understanding of how much liquidity banks create, how this liquidity creation changes over time, and the key factors that affect it. In this paper, we construct four comprehensive measures of bank liquidity creation and apply them to data on U.S. banks from 1993-2003. Based on our preferred measure, the amount of liquidity created by the banking sector has grown from less than \$1.0 trillion in 1993 to over \$1.5 trillion in 2003 in real dollars. Existing theories also provide specific predictions regarding the effect of bank capital on liquidity creation that have not been tested. We test the net effect of bank capital on liquidity creation to examine the circumstances under which the different theories empirically dominate. We find very different results for large and small banks. For large banks, capital has a weak positive effect on liquidity creation and for small banks, the effect is strongly negative. These results are robust to using alternative methods of measuring liquidity creation; using different measures of capital; and corrections to address a potential endogeneity problem associated with bank capital.

[†] Board of Governors of the Federal Reserve System and Wharton Financial Institutions Center. Contact details: Mail Stop 153, Federal Reserve Board, 20th and C Sts. N.W., Washington, D.C. 20551. Tel.: 202-452-2903. Fax: 202-452-5295. E-mail: aberger@frb.gov.

[‡] Weatherhead School of Management, Case Western Reserve University, 10900 Euclid Avenue, 362 Peter B. Lewis Building, Cleveland, OH 44106. Tel.: 216-368-3688. Fax: 216-368-6249. E-mail: christa.bouwman@case.edu.

Keywords: Capital Structure, Liquidity Creation, Regulation, and Banking.

JEL Classification: G32, G28, and G21.

The views expressed do not necessarily reflect those of the Federal Reserve Board or its staff. A previous version of this paper was entitled “Bank capital and liquidity creation.” The authors thank Bob Avery, Bill Bassett, Sreedhar Bharath, Arnoud Boot, Bob DeYoung, Doug Diamond, Phil Dybvig, Mark Flannery, Michael Gordy, Diana Hancock, Gautam Kaul, Lutz Kilian, Beth Kiser, Vikram Nanda, Charlotte Ostergaard, George Pennacchi, Jianping Qi, Rich Rosen, Doug Skinner, Anjan Thakor, Bent Vale, Egon Zakrajsek, and participants at presentations at the Western Finance Association Meeting, the Financial Intermediation Research Society Meeting, the Federal Reserve Bank of Chicago’s Bank Structure and Competition Conference, the University of Michigan, Case Western Reserve University, Ohio State University, the Federal Reserve Board, the Federal Reserve Bank of Cleveland, and the Norges Bank for helpful comments, and Phil Ostromogolsky for excellent research assistance. Part of this research was conducted while Christa Bouwman was visiting the Federal Reserve Board.

1. Introduction

Analyses of the role of banks in creating liquidity and thereby spurring economic growth have a long tradition, dating back to Adam Smith (1776).¹ Modern reincarnations of the idea that liquidity creation is central to banking appear most prominently in the formal analyses in Bryant (1980) and Diamond and Dybvig (1983). These theories argue that banks create liquidity on the balance sheet by financing less liquid assets with more liquid liabilities, an insight that is also closely related to the literature on financial intermediary existence (e.g., Diamond 1984, Ramakrishnan and Thakor 1984). Holmstrom and Tirole (1998) and Kashyap, Rajan, and Stein (2002) suggest that banks also create liquidity off the balance sheet through loan commitments and similar claims to liquid funds.

Although the importance of liquidity creation by banks for the economy and markets is generally well accepted, the striking absence of empirical measures of bank liquidity creation makes it difficult to assess the size or pervasiveness of this effect. That is, we do not know the magnitude of bank liquidity creation, the intertemporal behavior of bank liquidity creation, and the factors that affect bank liquidity creation. As a result, it is difficult to address numerous questions of research and policy interest. Does liquidity creation affect economic growth? How do monetary policy initiatives by central banks affect liquidity creation? How does liquidity creation differ across nations? How much liquidity do banks create relative to financial markets, and what are the complementarities, if any, in liquidity creation across banks and capital markets? What is the effect of bank capital on liquidity creation by banks?

We do not address all of these questions here – rather, our primary goal is to develop measures of bank liquidity creation that can be used to address these issues. We apply these measures to data on virtually all U.S. banks over 1993-2003, and thereby take an important first step into understanding how much liquidity banks create and how liquidity creation has changed over time. Our secondary goal is to

¹ Smith (book II, chapter II, 1776) highlights the importance of liquidity creation by banks and describes how it helped commerce in Scotland. In particular, he notes: “That the trade and industry of Scotland, however, have increased very considerably during this period, and that the banks have contributed a good deal to this increase, cannot be doubted.”

use our liquidity creation measures to study one of the policy-relevant research issues highlighted above – the effect of bank capital on bank liquidity creation. The existing theories produce specific predictions on this issue that are potentially testable but have yet to be confronted with the data. Some theories predict that capital reduces bank liquidity creation, while others predict that capital allows banks to create more liquidity. Because these theories arrive at their predictions by modeling entirely different economic forces, it is possible that their predictions apply in different circumstances. We develop economic intuition with respect to these circumstances and then test the effect of capital on liquidity creation.

To construct our liquidity creation measures, we use a three-step procedure. In Step 1, we classify all bank assets, liabilities, equity, and off-balance sheet activities as liquid, semi-liquid, and illiquid. We do this based on the ease, cost, and time for customers to obtain liquid funds from the bank, and the ease, cost, and time for banks to dispose of their obligations in order to meet these liquidity demands. Our use of just three liquidity classifications (liquid, semi-liquid, and illiquid) is a necessary simplification – any finer distinctions would have to be made rather arbitrarily. In Step 2, we assign weights to the activities classified in Step 1. The weights are consistent with the theory – maximum (i.e., dollar-for-dollar) liquidity is created when illiquid assets are transformed into liquid liabilities and maximum liquidity is destroyed when liquid assets are transformed into illiquid liabilities or equity. In Step 3, we construct four liquidity creation measures by combining the activities as classified in Step 1 and as weighted in Step 2 in different ways. The measures classify all activities other than loans by both product category and maturity but – due to data limitations – classify loans based either on category (“cat”) or on maturity (“mat”). We also alternatively include off-balance sheet activities (“fat”) or exclude them (“nonfat”). We thus construct liquidity creation measures based on the four combinations, “cat fat,” “mat fat,” “cat nonfat,” and “mat nonfat.” As explained in more detail below, “cat fat” is our preferred measure.

Our calculations suggest that banks created over \$1.5 trillion in liquidity in 2003 using our

preferred “cat fat” measure.² This is approximately equal to 22% of bank gross total assets or GTA (total assets plus allowance for loan and lease losses and the allocated transfer risk reserve) and about two and a half times the overall level of bank equity capital. Thus, by our estimates, the industry creates approximately \$2.5 of liquidity per \$1 of capital.

Liquidity creation has grown dramatically over time. Using our preferred “cat fat” measure, overall bank liquidity creation grew by approximately two-thirds from less than \$1.0 trillion in 1993 to over \$1.5 trillion in 2003 in real terms. Our results are fairly similar when we calculate liquidity creation using our “mat fat” measure that classifies loans based on maturity instead of category. When we use our “nonfat” measures instead – which exclude off-balance sheet activities – the patterns of liquidity creation over time are similar, but the dollar amounts of liquidity creation are much lower. These findings highlight the importance of liquidity created off the balance sheet.

Liquidity creation differs dramatically between large banks (GTA exceeding \$1 billion) and small banks (GTA up to \$1 billion) (measured in real 2003 dollars). Based on our preferred “cat fat” measure, large banks are responsible for 85% of industry liquidity creation and about 80% of industry assets, while comprising only about 5% of the sample observations. Both size classes generate significant portions of their liquidity off the balance sheet, but the fraction of liquidity created by loan commitments and similar claims to liquid funds is much higher for large banks. Both size classes increased liquidity creation in real terms over the sample period, but perhaps surprisingly, small banks had greater growth.

Turning to the theories on the relationship between bank capital and liquidity creation, some recent contributions suggest that bank capital may impede liquidity creation by making the bank’s capital structure less fragile (e.g., Diamond and Rajan 2000, 2001). A fragile capital structure encourages the bank to commit to monitoring its borrowers, and hence allows it to extend loans. Additional equity capital makes it harder for the less-fragile bank to commit to monitoring, which in turn hampers the bank’s ability to create liquidity. Capital may also reduce liquidity creation because it

² All liquidity creation measures in the paper are as of December 31 of a given year.

“crowds out” deposits (e.g., Gorton and Winton 2000). For expositional ease, we refer to this first set of theories jointly as the “financial fragility-crowding out” hypothesis of capital.

An alternative view is that higher capital improves banks’ ability to absorb risk and hence create liquidity. Liquidity creation exposes banks to risk – the more liquidity is created, the greater are the likelihood and severity of losses associated with having to dispose of illiquid assets to meet customers’ liquidity demands (Diamond and Dybvig 1983, Allen and Santomero 1998, Allen and Gale 2003). A well-known role of capital is to absorb risk and expand banks’ risk-bearing capacity (e.g., Bhattacharya and Thakor 1993, Repullo 2004, Von Thadden 2004), so higher capital ratios may allow banks to create more liquidity. We refer to this second set of theories collectively as the “risk absorption” hypothesis, while recognizing that the theories together rather than separately produce this prediction.

Both the “financial fragility-crowding out” and the “risk absorption” effects may apply in differing degrees to liquidity creation by different banks, so the relevant empirical issue is discovering the circumstances under which each effect empirically dominates. We address this issue by testing whether the net effect of bank capital on liquidity creation is negative or positive, and by examining whether this net effect differs across types of banks in some intuitively predictable ways. Specifically, we expect that the “financial fragility-crowding out” effect is likely to be relatively strong for small banks. One reason is that, compared with large banks, small banks deal more with entrepreneurial-type small businesses, where the close monitoring highlighted in Diamond and Rajan (2000, 2001) is important. A second reason is that small banks tend to raise funds locally, so that capital may “crowd out” deposits, whereas this effect is likely to be relatively weak for large banks that can more easily access funding from national or international capital markets. In contrast, the “risk absorption” effect is likely to be stronger for large banks because they are generally subject to greater regulatory scrutiny and market discipline than small banks, which may affect their capacity to absorb risk. The relationship between bank capital and liquidity creation may also depend on the level of bank capital. We expect that for both size classes, the “risk absorption” effect may be relatively strong for banks with low capital ratios because these banks have the smallest buffers to absorb risks and face greater regulatory, market,

and/or owner pressures to control risk taking.

To test the net effect of capital on liquidity creation, we regress the dollar amount of bank liquidity creation (calculated using our four measures and normalized by GTA) for each bank-year observation on the bank's lagged equity capital ratio and a number of control variables. We use three-year lagged average values of capital and the other exogenous variables to mitigate potential endogeneity problems, as lagged values represent earlier bank decisions. We run the tests separately for large and small banks and also for banks in both size classes split by the level of the lagged capital ratio to allow for the possibility that capital may affect these banks differently.

We find empirical support for both hypotheses that is generally consistent with economic intuition. For large banks, capital has a weak positive effect on liquidity, consistent with the expected empirical dominance of the "risk absorption" effect. In sharp contrast, for small banks, the effect of capital on liquidity creation is strongly negative, consistent with the expected dominance of the "financial fragility-crowding out" effect for these institutions. To understand more deeply why these differences exist, we also examine the impact of capital on the individual components of liquidity creation for large and small banks.

When we split large and small banks by their lagged capital ratios into low-, medium-, and high-capital thirds and rerun our regressions, we also obtain results consistent with economic intuition. We find that capital has a more positive effect on liquidity creation for low-capital banks than for higher-capital banks in both size classes. This is consistent with the expected stronger risk-absorption effect for banks with low capital.

We show that our regression results are generally robust to using alternative methods to measuring liquidity creation and to using different measures of capital. We also address in depth the potential endogeneity of capital in our regressions. We recognize that capital and liquidity creation may be jointly determined, and that our use of three-year lagged average values of capital may not be sufficient to mitigate such endogeneity concerns. We therefore also use an instrumental variable approach with four instruments for lagged capital (bank holding company dummy, dividend payer

dummy, effective state income tax rate, lagged loan loss provisions) and find that our main results are qualitatively unaffected.

The remainder of the paper is organized as follows. Section 2 reviews the literature. Section 3 describes the construction of our liquidity creation measures. Section 4 discusses our panel data set of U.S. banks over 1993-2003 and shows how our four measures of liquidity creation vary over this time period. Section 5 outlines our regression framework and Section 6 contains our regression results. Section 7 addresses robustness issues, and Section 8 concludes.

2. Literature review

In this section, we provide a brief review of the literature to place our paper in context. Our research is related to three strands of literature: the measurement of bank liquidity creation; the theories of the effect of capital on liquidity creation; and the empirical studies of capital and liquidity creation. We discuss these three literatures in turn.

2.1. Measurement of bank liquidity creation

We are aware of only one paper that measures bank liquidity creation. Deep and Schaefer (2004) construct a measure of liquidity transformation and apply it to data on the 200 largest U.S. banks from 1997 to 2001. They define the liquidity transformation gap or “LT gap” as $(\text{liquid liabilities} - \text{liquid assets}) / \text{total assets}$. They consider all loans with maturity of one year or less to be liquid, and they explicitly exclude loan commitments and other off-balance sheet activities because of their contingent nature. They find that the LT gap is about 20% of total assets on average for their sample of large banks. The authors conclude that these banks do not appear to create much liquidity, and run some tests to explain this finding, examining the roles of insured deposits, credit risk, and loan commitments.

The LT gap is an intuitive step forward, but we do not believe it is sufficiently comprehensive. To illustrate, we highlight a few key differences between their approach and ours. First, we include virtually all commercial banks and compare findings for large and small banks, rather than including only the largest institutions. Second, our preferred “cat fat” liquidity creation measure classifies loans

by category (“cat”), rather than by maturity. We treat business loans as illiquid because banks generally cannot easily dispose of them to meet liquidity needs, but we treat residential mortgages and consumer loans as semi-liquid because these loans can often be securitized and sold to meet demands for liquid funds. Third, our preferred measure includes off-balance sheet activities (“fat”), consistent with the arguments in Kashyap, Rajan, and Stein (2002) and others. In our less-preferred liquidity measures, we try classifying loans by maturity (“mat”) and excluding off-balance sheet activities (“nonfat”) to determine the effects of these assumptions. As discussed in Section 3 below, the LT gap is conceptually close to our “mat nonfat” measure.

2.2. Bank liquidity creation and capital: the theories

As noted in the Introduction, the theoretical literature produces opposing predictions on the link between capital and liquidity creation. One set of theories – which we refer to collectively as the “financial fragility-crowding out” hypothesis – predicts that higher capital reduces liquidity creation. Diamond and Rajan (2000, 2001) focus on financial fragility. They model a relationship bank that raises funds from investors to provide financing to an entrepreneur. The entrepreneur may withhold effort, which reduces the amount of bank financing attainable. More importantly, the bank may also withhold effort, which limits the bank’s ability to raise financing. A deposit contract mitigates the bank’s holdup problem – because depositors can run on the bank if the bank threatens to withhold effort – and therefore maximizes liquidity creation. Providers of capital cannot run on the bank, which limits their willingness to provide funds, and hence reduces liquidity creation. Thus, the higher a bank’s capital ratio, the less liquidity it will create.³

Gorton and Winton (2000) show how a higher capital ratio may reduce liquidity creation through the crowding out of deposits. They argue that deposits are more effective liquidity hedges for investors than investments in equity capital. Thus, higher capital ratios shift investors’ funds from

³ Diamond and Rajan’s model builds on Calomiris and Kahn’s (1991) argument that the ability of uninsured depositors to run on the bank in the event of expected wealth expropriation by bank managers is an important disciplining mechanism. A related idea is proposed by Flannery (1994), who focuses on the disciplining effect of depositors’ ability to withdraw funds on demand, and thus prevent the bank from expropriating depositor wealth through excessively risky investments.

relatively liquid deposits to relatively illiquid bank capital, reducing overall liquidity for investors.⁴

Under the alternative “risk absorption” hypothesis, higher capital enhances banks’ ability to create liquidity. This insight is based on two strands of the literature. One strand consists of papers (e.g., Diamond and Dybvig 1983, Allen and Santomero 1998, Allen and Gale 2004) that argue that liquidity creation exposes banks to risk. The more liquidity that is created, the greater is the likelihood and severity of losses associated with having to dispose of illiquid assets to meet the liquidity demands of customers.

The second strand consists of papers (e.g., Bhattacharya and Thakor 1993, Repullo 2004, Von Thadden 2004) that posit that bank capital absorbs risk and expands banks’ risk-bearing capacity. Combining these two strands yields the prediction that higher capital ratios may allow banks to create more liquidity.

Finally, we point out one additional contribution of some of the recent theories. The standard view of liquidity creation is that banks create liquidity by transforming illiquid assets into liquid liabilities. Diamond and Rajan (2000, 2001) and Gorton and Winton (2000) show, however, that banks can create more or less liquidity by simply changing their funding mix on the liability side. Thakor (1996) shows that capital may also affect banks’ asset portfolio composition, thereby affecting liquidity creation through a change in the asset mix. Our measures of liquidity creation incorporate these insights – they explicitly recognize that liquidity creation by banks occurs through changes in the mixes on both sides of the balance sheet as through off-balance sheet activities.

2.3. Bank capital and liquidity creation: the existing empirical evidence

Some empirical studies examine issues related to liquidity creation, but do not focus on the role of capital. Kashyap, Rajan, and Stein (2002) provide empirical evidence of synergies between commitment lending and deposits, consistent with their model. Gatev, Schuermann, and Strahan (2005) and Gatev and Strahan (forthcoming) find that banks have a comparative advantage in hedging liquidity

⁴ Gorton and Winton’s (2000) analysis suggests that even if equity holders did not reduce funding to the bank in Diamond-Rajan (2000), there would be less liquidity creation with a higher capital ratio.

risk in the economy because banks experience deposit inflows following a market crisis or liquidity shock that allow them to have more funds to provide the additional loans drawn down under commitments at such times. Pennacchi (2006) confirms the existence of synergies between loan commitments and deposit taking, but finds that such synergies do not hold prior to the creation of FDIC deposit insurance. These studies do not focus on the role of bank capital, but they do in some cases include the capital ratios in regressions of some liquidity creation components, yielding ambiguous predictions related to the effect of capital on liquidity creation.⁵

The credit crunch literature tests hypotheses about bank capital and one type of liquidity creation, usually business lending or real estate lending, during the early 1990s when bank lending declined significantly. Several studies find that the decline in bank capital ratios arising from loan losses in the late 1980s and early 1990s contributed significantly to the reduction in lending (e.g., Peek and Rosengren 1995). This is consistent with a positive relationship between capital and liquidity creation during a period of distress. In the early 1990s, U.S. regulators also imposed new leverage requirements, as well as the Basel I risk-based capital standards. Most of the studies found that the leverage requirements contributed to the decline in lending, again consistent with the hypothesis of a positive effect of bank capital on liquidity creation (e.g., Berger and Udell 1994, Hancock, Laing, and Wilcox 1995, Peek and Rosengren 1995). However, most studies of the Basel I standards found little effect on lending. Unfortunately, the unusual combination of several major changes in bank capital regulation and a recession makes it difficult to parse the different effects and draw general conclusions.

Finally, some studies of bank lending behavior include capital ratios, but focus on other issues. For example, Berger and Udell (2004) study procyclical lending and find positive, statistically significant effects of capital on the annual growth of business loans. Holod and Peek (2004) examine monetary policy effects and find that the capital ratio has significant positive effects on loan growth. Gambacorta and Mistrulli (2004) use Italian data and find that the impact of monetary policy and GDP

⁵ For example, Gatev and Strahan find that a higher bank capital ratio tends to be followed by greater loans and deposits (which may increase liquidity creation) and greater liquid assets and non-deposit liabilities (which may reduce liquidity creation).

shocks on bank lending depends on bank capitalization.

Thus, the existing empirical literature sheds relatively little light on the relationship between bank capital and liquidity creation. Some of the studies test the liquidity creation theories, but do not focus on the role of bank capital. Others include capital in regressions, but specify only limited components of liquidity creation, and often under unusual circumstances. Our empirical analysis uses a significantly different approach.

3. Construction of our liquidity creation measures

In this section, we explain the construction of our four liquidity creation measures. We clarify which of these measures is our preferred measure. We also show how Deep and Schaefer's (2004) liquidity transformation measure can be viewed as a special case of one of our measures.

We construct the liquidity creation measures using a three-step procedure. In Step 1, we classify all bank balance sheet and off-balance sheet activities as liquid, semi-liquid, or illiquid. In Step 2, we assign weights to the activities classified in Step 1. In Step 3, we combine the activities as classified in Step 1 and as weighted in Step 2 in different ways to construct our four liquidity creation measures, "cat fat," "mat fat," "cat nonfat," and "mat nonfat." The first page of Table 1 illustrates Steps 1 and 2 and the second page of Table 1 illustrates Step 3.

3.1. Step 1 – Classifying activities as liquid, semi-liquid, or illiquid

In Step 1, we classify all assets as liquid, semi-liquid, or illiquid based on the ease, cost, and time for banks to dispose of their obligations to obtain liquid funds to meet customers' demands. We similarly classify bank liabilities plus equity as liquid, semi-liquid, or illiquid, based on the ease, cost, and time for customers to obtain liquid funds from the bank. Off-balance sheet guarantees and derivatives are classified consistently with treatments of functionally similar on-balance sheet items.⁶

Ideally, we would use information on both product category and maturity to classify all bank

⁶ In a robustness check, we use an alternative approach to measuring the liquidity contribution of some of these items (see Section 7.1).

activities. For example, as noted above, business loans are generally more illiquid than residential mortgages and consumer loans, as the latter can often be more easily securitized and sold to meet liquidity demands. Within each category, shorter maturity items are more liquid than longer maturity items because they self-liquidate without effort or cost sooner.

For bank activities other than loans, Call Reports provide sufficient detail on category and maturity, so our classifications incorporate both aspects. Unfortunately, this is not the case for loans. Call Reports split loans into various loan categories and into different maturity classes, but do not provide maturity information for individual loan categories. We therefore either classify loans entirely by category (“cat”) or entirely by maturity (“mat”). Thus, our “cat” and “mat” liquidity creation measures constructed below classify loans either by category or maturity, but in all cases incorporate both key characteristics for other bank activities.

Assets

- Classifying loans:
 - *category (“cat”)*: For the “cat” measures of liquidity creation, we classify business loans and leases as illiquid assets, because these items typically can not be sold quickly without incurring a major loss. Residential mortgages and consumer loans are generally relatively easy to securitize, and loans to depositories and governments are likely to be comparatively easy to sell or otherwise disposed of because the counterparties are relatively large and informationally transparent. We classify these loan categories as semi-liquid assets.⁷
 - *maturity (“mat”)*: As discussed above, shorter maturity items are more liquid than longer maturity items because they self-liquidate sooner. We therefore classify all short-term loans of up to one year as semi-liquid and all long-term loans of over one year as illiquid for the “mat” measures.
- Classifying assets other than loans: We classify premises and investments in unconsolidated

⁷ In a robustness check, we use a different method to establishing which loans are securitizable (see Section 7.2).

subsidiaries as illiquid assets, because typically these items can not be sold quickly without incurring a major loss. We classify cash, securities, and other marketable assets that the bank can use to meet liquidity needs quickly without incurring major losses as liquid assets.

Liabilities plus equity

- **Classifying liabilities:** We count funds that can be quickly withdrawn by customers – such as transactions deposits and overnight federal funds purchased – as liquid liabilities. We classify other deposits that can be withdrawn with slightly more difficulty or penalty as semi-liquid. We also classify as semi-liquid the balance sheet item ‘other borrowed money,’ which contains other short- and medium-maturities with terms longer than overnight, such as term federal funds, repurchase agreements, and borrowings from Federal Reserve Banks and Federal Home Loan Banks. We classify liabilities that generally cannot be withdrawn easily or quickly, such as subordinated debt, as illiquid.
- **Classifying equity:** We classify equity as illiquid because investors can not demand liquid funds from the bank and the maturity is very long. Although the equity of some banks is publicly traded and may be sold relatively easily, the investors are able to retrieve liquid funds through the capital market, not from the bank. Thus, while traded equity may be liquid from an individual investor’s point of view, such liquidity is created by the capital market, rather than by the bank, the focus of this paper.

Off-balance sheet activities

- **Classifying guarantees:** We classify loan commitments and commercial letters of credit as illiquid guarantees. These items are functionally similar to on-balance sheet business loans in that they are obligations that are illiquid from the point of view of the bank – except in very unusual circumstances, the bank must provide the funds to the customer upon demand.⁸ As well, the bank

⁸ We acknowledge that banks could dispose of loan commitments by invoking the material adverse change (MAC) clause and the customer would not have access to the funds. However, failing to honor loan commitments is generally very costly since it may create legal liabilities and reputational losses, and is therefore rarely done.

generally cannot sell or participate these items. We classify net standby letters of credit (i.e., the amount outstanding minus the amount conveyed to others), net credit derivatives (i.e., the amount guaranteed minus the beneficiary amount), and net securities lent (i.e., the amount lent minus the amount borrowed) as semi-liquid guarantees since they can potentially be sold or participated, analogous to semi-liquid on-balance sheet residential mortgages and consumer loans. We classify net participations acquired from other institutions (i.e., the amount acquired minus the amount conveyed to others) as liquid guarantees, since they are functionally similar to on-balance sheet securities.

- Classifying derivatives: We classify all derivatives – interest rate, foreign exchange, and equity and commodity derivatives – as liquid because they can be bought and sold easily and are functionally similar to liquid securities. We focus on the fair values of these derivatives, which measure how much liquidity the bank is providing to or absorbing from the public.

3.2. Step 2 – Assigning weights to the activities classified in Step 1

In Step 2, we assign weights to all of the bank activities classified in Step 1. That is, we assign weights to the classes of liquid, semi-liquid, and illiquid assets, liabilities plus equity, and off-balance sheet guarantees and derivatives shown on the first page of Table 1.

We base the weights on liquidity creation theory. According to this theory, banks create liquidity on the balance sheet when they transform illiquid assets into liquid liabilities. An intuition for this is that banks create liquidity because they hold illiquid items in place of the nonbank public and give the public liquid items. We therefore apply positive weights to both illiquid assets and liquid liabilities, so when liquid liabilities – such as transactions deposits – are used to finance illiquid assets – such as business loans – liquidity is created. Following similar logic, we apply negative weights to liquid assets, illiquid liabilities, and equity, so that when illiquid liabilities or equity is used to finance a dollar of liquid assets – such as treasury securities – liquidity is destroyed.

The magnitudes of the weights are based on simple dollar-for-dollar adding-up constraints, so that \$1 of liquidity is created when banks transform \$1 of illiquid assets into \$1 of liquid liabilities.

Similarly, we require that \$1 of liquidity is destroyed when banks transform \$1 of liquid assets into \$1 of illiquid liabilities. Based on these constraints, we assign a weight of $\frac{1}{2}$ to both illiquid assets and liquid liabilities and a weight of $-\frac{1}{2}$ to both liquid assets and illiquid liabilities. Thus, when a dollar of liquid liabilities – such as transactions deposits – is used to finance a dollar of illiquid assets – such as business loans – liquidity creation equals $\frac{1}{2} * \$1 + \frac{1}{2} * \$1 = \$1$. In this case, maximum liquidity (\$1) is created. Intuitively, the weight of $\frac{1}{2}$ applies to both illiquid assets and liquid liabilities, since the amount of liquidity created is only “half” determined by the source or use of the funds alone – both are needed to create liquidity. Similarly, when a dollar of illiquid liabilities or equity is used to finance a dollar of liquid assets – such as treasury securities – liquidity creation equals $-\frac{1}{2} * \$1 + -\frac{1}{2} * \$1 = -\$1$, as maximum liquidity is destroyed.

Using these weights, banks do not create liquidity when they use liquid liabilities (e.g., transaction deposits) to finance liquid assets (e.g., treasuries), or when they use illiquid liabilities or equity to finance illiquid assets (e.g., business loans). In these cases, banks hold items of approximately the same liquidity as they give to the nonbank public.

We apply the intermediate weight of 0 to semi-liquid assets and liabilities, based on the assumption that semi-liquid activities fall halfway between liquid and illiquid activities. Thus, the use of time deposits to fund residential mortgages would yield approximately zero net liquidity creation, since the ease, cost, and time with which the time depositors may access their funds early and demand liquidity roughly equals the ease, cost, and time with which the bank can securitize and sell the mortgage to provide the funds.

We apply weights to off-balance sheet guarantees and derivatives using the same principles, consistent with the functional similarities to on-balance sheet items discussed in Step 1. For example, illiquid off-balance sheet guarantees – such as loan commitments – are functionally similar to on-balance sheet illiquid loans – such as business loans – in that they are obligations of the bank to provide funds that cannot be easily sold or participated. We therefore apply the same weight of $\frac{1}{2}$ to illiquid guarantees as we do to illiquid assets. Similarly, we apply the same weight of 0 to semi-liquid

guarantees as we do to functionally similar semi-liquid on-balance sheet assets, and we apply the same weight of $-\frac{1}{2}$ to liquid guarantees that we do to functionally similar on-balance sheet liquid assets.

Analogously, the gross fair values of derivatives are assigned the same weight of $-\frac{1}{2}$ as on-balance sheet liquid assets. As discussed in Step 1, these contracts can be bought and sold easily and are functionally similar to liquid securities. The Call Reports separate these contracts into those with gross positive and gross negative fair values.⁹ For derivatives with gross positive fair values, this reduces bank liquidity creation as the bank effectively holds a valuable liquid asset in place of the public. For derivatives with gross negative fair values, the negative value is multiplied by a weight of $-\frac{1}{2}$, yielding a positive contribution to bank liquidity creation as the bank effectively holds a negatively-valued liquid asset in place of the public.

We arrange the columns on the first page of Table 1 such that all the bank activities that contribute to liquidity creation are on the left, all those that subtract from liquidity creation are on the right, and all those with an approximately neutral effect on liquidity creation are in the center. Thus, those that are assigned a weight of $\frac{1}{2}$ – illiquid assets, liquid liabilities, and illiquid guarantees – are grouped together on the left. Liquid assets, illiquid liabilities plus equity, and liquid guarantees and derivatives – which are assigned a weight of $-\frac{1}{2}$ – are grouped on the right. Finally, semi-liquid assets, liabilities, and guarantees with 0 weights are grouped in the center.

3.3. Step 3 – Constructing liquidity creation measures by combining activities as classified in Step 1 and as weighted in Step 2

In Step 3, we combine the activities as classified and weighted in Step 1 and Step 2, respectively, in different ways to construct our liquidity creation measures. The measures are similar in that they all classify activities other than loans using information on product category and maturity, as discussed in Step 1. The measures differ in that we alternatively classify loans by category or maturity (“cat” versus “mat”), and alternatively include or exclude off-balance sheet activities (“fat” versus “nonfat”). Hence,

⁹ Fair values are as in FASB 133: the amount at which an asset (liability) could be bought (incurred) or sold (settled) in a current transaction between willing parties, that is, other than in a forced or liquidation sale. The fair value equals the quoted market price, if available. If a quoted market price is not available, the estimate of fair value is based on the best information available in the circumstances.

we have four measures: “cat fat,” “cat nonfat,” “mat fat,” and “mat nonfat.” The formulas are shown on the second page of Table 1. On that page, we again arrange the bank activities that add to liquidity creation on the left, those that subtract from liquidity creation on the right, and those with an approximately neutral effect in the center. For all measures, we multiply the weights of $\frac{1}{2}$, $-\frac{1}{2}$, or 0, respectively, times the dollar amounts of the corresponding bank activities to arrive at the total dollar value of liquidity creation at a particular bank. We sum across all banks to obtain the total dollar value of liquidity created by the entire industry.

We recognize that our liquidity creation measures are rough approximations. We classify all bank activities as liquid, semi-liquid, or illiquid, and use three weights, $\frac{1}{2}$, 0, and $-\frac{1}{2}$. Differences in liquidity obviously exist within each of the three classifications, but the data generally do not allow for much finer distinctions, and there are no other unambiguous weights to apply. The use of $\frac{1}{2}$, $-\frac{1}{2}$, and 0 are the clear demarcations of full liquidity, full illiquidity, and neutrality, respectively, and no other clear choices present themselves.

It is interesting to note that Deep and Schaefer’s (2004) LT gap measure is conceptually close to our “mat nonfat” measure and may be viewed as a special case of it. If we classified all assets and liabilities as either liquid or illiquid (none as semi-liquid) using maturities, excluded off-balance sheet activities, and specified assets (A) rather than gross total assets (GTA), our “mat nonfat” formula reduces to their formula.¹⁰

We next discuss why we consider “cat fat” to be our preferred liquidity creation measure. First, we argue that the “cat” measures are preferred to the “mat” measures primarily because what matters to liquidity creation on the asset side is the ease, cost, and time for banks to dispose of their obligations to obtain liquid funds. The ability to securitize loans is closer to this concept than the time until self-liquidation – for example, a 30-year residential mortgage may be securitized relatively quickly even though it is a long-term loan. We also argue that the “fat” measures are preferred to the “nonfat”

¹⁰ Applying these changes, our formula becomes $[\frac{1}{2} \cdot \text{illiquid assets} - \frac{1}{2} \cdot \text{liquid assets} + \frac{1}{2} \cdot \text{liquid liabilities} - \frac{1}{2} \cdot \text{illiquid liabilities} - \frac{1}{2} \cdot \text{equity}] / A = [\frac{1}{2} \cdot (A - \text{liquid assets}) - \frac{1}{2} \cdot \text{liquid assets} + \frac{1}{2} \cdot (\text{liquid liabilities}) - \frac{1}{2} \cdot (\text{assets} - \text{liquid liabilities})] / A = [\text{liquid liabilities} - \text{liquid assets}] / A$, which is their LT gap measure.

measures based on the argument that off-balance sheet activities provide liquidity in functionally similar ways to on-balance sheet items. Hence, “cat fat” is our preferred measure.

4. Sample description and bank liquidity creation over time

In this section, we first describe how we construct our sample. We then explain how bank liquidity creation has changed over our sample period using our four measures.

4.1. Sample description

Our sample includes annual data as of December 31 on almost all commercial banks from 1993 to 2003. To ensure that our sample only contains ‘true,’ viable commercial banks, we impose the following restrictions. We exclude a bank if it: 1) has no loans outstanding; 2) has no commercial real estate or commercial and industrial loans outstanding; 3) has zero deposits; 4) has zero or negative equity capital in the current or lagged year; 5) is very small (average lagged GTA below \$25 million);¹¹ 6) has unused commitments exceeding four times GTA; 7) resembles a thrift (residential real estate loans exceeding 50% of GTA); or 8) is classified by the Federal Reserve as a credit card bank or has consumer loans exceeding 50% of GTA.¹² We also eliminate 0.7% of all bank-year observations because some of the exogenous variables used in our regression analysis are missing. Our sample contains 84,080 bank-year observations: 3,937 bank-year observations for large banks (GTA exceeding \$1 billion), and 80,143 for small banks (GTA up to \$1 billion).¹³

4.2. Liquidity creation over time

Table 2 shows summary statistics on our four liquidity creation measures for the entire banking sector, and separately for large and small banks. Panel A shows liquidity creation in 1993 and in 2003 – the

¹¹ Banks with lagged average GTA below \$25 million are not likely to be viable commercial banks in equilibrium. This exclusion reduced the sample size by 12,880 bank-year observations (from 96,960 to 84,080), but does not materially affect our findings. Inclusion of these banks increases liquidity creation of small banks by only 0.6% (\$0.0013 trillion) in 2003 based on our “cat fat” measure, and leaves our regression results qualitatively unchanged.

¹² The Federal Reserve Board defines a credit card bank as having: 1) 50% or more of its total assets in the form of loans to individuals; 2) 90% or more of its loans to individuals in the form of credit card outstandings; and 3) \$200 million or more in loans to individuals.

¹³ We apply the \$1 billion cutoff, measured in real 2003 dollars, in each year to separate banks in our sample into large and small banks.

first and last years of our sample period, respectively. Panel B shows graphs of the corresponding liquidity creation measure over the entire sample period.

Starting with our preferred “cat fat” measure – which classifies loans by category and includes off-balance sheet activities – we find that banks created liquidity of \$1.570 trillion in 2003 (see Table 2 Panel A). This equals 22% of industry GTA (shown) and represents about \$2.5 of liquidity per \$1 of equity capital (not shown). Overall liquidity creation increased by approximately two-thirds in real terms from 1993 to 2003. As shown, liquidity creation as a fraction of GTA increased by just one percentage point from 0.21 to 0.22, so liquidity creation grew at a slightly faster rate than GTA. At large banks, liquidity creation increased by almost 60% in real terms, although it declined as a fraction of GTA from 26% in 1993 to 22% in 2003. Large banks created approximately 85% of industry liquidity at the end of the sample period. Perhaps surprisingly, small banks experienced a much larger percentage increase in liquidity creation over the sample period – more than a 100% increase both in real terms and as a fraction of GTA – but still accounted for only about 15% of industry liquidity creation in 2003.

As shown in Table 2 Panel B, despite the significant increase in measured liquidity creation over our sample period based on our “cat fat” measure, there is a slight decline over the final three years. Analysis of the individual components of liquidity creation over time (not shown in the interest of brevity) suggests that three components of liquidity creation in particular are responsible for this pattern: 1) illiquid assets; 2) illiquid off-balance sheet guarantees; and 3) liquid assets. The major growth in liquidity creation from 1993-2000 was largely driven by strong increases in illiquid assets (weight = $\frac{1}{2}$) and illiquid off-balance sheet guarantees (weight = $\frac{1}{2}$) that outweighed the smaller increases in liquid assets (weight = $-\frac{1}{2}$). Over 2000-2003, illiquid assets showed virtually no growth, illiquid guarantees increased slightly, while the growth in liquid assets continued, causing a mild decline in liquidity creation in these final three years. Note that this pattern occurred only for large banks, which create the majority of liquidity of the banking sector. Liquidity creation by small banks increased in virtually every year over our sample period.

Turning briefly to the other liquidity creation measures, the data show much less measured liquidity creation for the “cat nonfat” measure – which is the same as “cat fat” except for the exclusion of off-balance sheet activities. In 1993, measured “cat nonfat” liquidity creation is less than 40% of that of “cat fat” and only 8% of GTA, and actually falls to only 5% of GTA by 2003. The drop in liquidity creation occurs entirely for large banks – small banks actually tripled their liquidity creation based on the “nonfat” measure – reflecting the much greater role that off-balance sheet activities play for large banks. The “cat nonfat” graph (see Panel B) shows that the decline in liquidity creation for this measure was concentrated in the final three years of the sample period, when liquidity creation by large banks fell considerably. Measured liquidity creation increased over 1993-2000 primarily because the above-mentioned increase in illiquid assets outweighed the increase in liquid assets, but the growth was less pronounced than for “cat fat” because the significant rise in illiquid off-balance sheet guarantees is not included in the “nonfat” measure. The decline over 2000-2003 is also more severe because the growth in liquid assets was not partially offset by an increase in illiquid off-balance sheet guarantees. All of these differences between the “cat fat” and “cat nonfat” measures highlight the importance of including off-balance sheet activities.

Liquidity creation is the highest in all years using our “mat fat” measure, which differs from our preferred “cat fat” measure by using loan maturities in place of categories to classify loans. Treating all loans with maturity of at least one year as illiquid assets increases measured liquidity creation substantially from 22% of GTA in 1993 to 27% in 2003, primarily because most residential mortgages are classified as illiquid (weight = $\frac{1}{2}$) rather than semi-liquid (weight = 0). The “mat fat” pattern of liquidity creation over time is similar to the “cat fat” pattern (see graphs in Panel B), except that liquidity creation picked up again in the final two years of the sample. The reason is that when loans are classified by maturity, illiquid assets (primarily residential mortgages) grew in those years, and their growth more than offset the growth in liquid assets.

The “mat nonfat” measure – which uses loan maturities and excludes off-balance sheet activities – yields a much smaller measured liquidity creation and its growth rate is below the growth

rate of GTA (liquidity creation drops from 12% of GTA in 1993 to 10% of GTA in 2003). The “mat nonfat” liquidity creation pattern resembles the “cat nonfat” pattern, except for the above-mentioned increase in liquidity creation in the final two years of the sample period.

5. Regression framework

We next turn to our second goal – to analyze the effect of bank capital on liquidity creation. In this section we discuss our regression framework, and in the next section, we present our regression results.

To examine whether the “financial fragility-crowding out” effect versus the “risk absorption” effect empirically dominates, we use panel data sets on large and small banks from 1993-2003. We regress the dollar amount of bank liquidity creation (calculated using our four liquidity creation measures) divided by GTA on the lagged equity capital ratio while controlling for other factors that may affect bank liquidity creation. Normalization by GTA is necessary to make the dependent variables meaningful and comparable across banks and to avoid giving undue weight to the largest institutions. Use of dollar amounts of liquidity creation without normalization would primarily amount to a regression of bank size on capital and other exogenous variables because banks differ so greatly in size even within the large-bank and small-bank size classes.

Our control variables include bank size, merger and acquisition history, and local market competition and economic environment, as explained in detail below. We include bank fixed effects to account for average differences over time across banks that are not captured by the other exogenous variables and to reduce correlations across error terms. Time fixed effects are added to control for average differences in liquidity creation across years that are not captured by the other exogenous variables, and to reduce serial correlation problems. All regressions are estimated with robust standard errors, clustered by bank, to control for heteroskedasticity as well as possible correlation among observations of the same bank in different years.

Table 3 gives descriptions and summary statistics for the exogenous variables. All financial values are expressed in real 2003 dollars using the implicit GDP price deflator. All of the exogenous

variables are lagged values averaged over the three years prior to observation of the dependent variables to reduce potential endogeneity problems, as lagged values are more likely to reflect earlier decisions. The use of three-year averages, rather than a single lagged year also reduces the effects of short-term fluctuations and problems with the use of accounting data. As well, portfolio changes take time to occur and likely reflect decisions made on the basis of historical experience, so three years of data may more accurately reflect the inputs into liquidity creation decisions. All of the lagged values are merger-adjusted – we collect information from the Federal Reserve Board’s National Information Center (NIC) database on a bank’s prior mergers and acquisitions, and use it to construct historical pro forma values.

The key exogenous variable is the lagged capital ratio. For our main analysis, we use EQRAT, the ratio of equity to GTA. Equity meets the most straightforward, narrow definition of capital as funds that cannot be easily withdrawn. GTA is the simplest measure of bank size, although it excludes off-balance sheet activities. GTA equals total assets plus the allowance for loan and lease losses and the allocated transfer risk reserve – two reserves held for potential credit losses – so that the full value of the loans financed and liquidity created by the bank on the asset side are included. In Section 7, we perform two robustness checks on the capital ratio. First, realizing that to some extent, a bank chooses its capital ratio, we use an instrumental variable approach to resolve any potential endogeneity problems. Second, we replace EQRAT with an alternative capital ratio.

To control for bank size, we include the natural log of bank size, $\ln(\text{GTA})$, in every regression, as well as running the regressions separately for large and small banks. The natural log is used for all of the continuous exogenous variables that may take on large values to avoid potential specification distortions, given that the dependent variables are virtually all in the $[0,1]$ interval.

We also control for the bank’s merger and acquisition history. The D-BANK-MERGE and D-DELTA-OWN dummies indicate whether a bank was involved in a merger or acquisition over the past three years, where a merger is defined as the combination of bank charters into an institution with a single set of books, and an acquisition is defined as a case in which the bank’s top-tier holding company changed with no change in charter status. Controlling for mergers and acquisitions is important because

banks often significantly alter their lending behavior following such events.

To construct controls for local market competition and economic environment, we define the local market as the Metropolitan Statistical Area (MSA) or non-MSA county in which the offices are located.¹⁴ For banks with offices in more than one local market, we use the weighted averages across these markets, using the proportion of the bank's deposits in each of these markets as the weights.¹⁵ To control for local market competition, we include HERF, the Herfindahl index of concentration for the market or markets in which the bank is present. We base HERF on the market shares of both banks and thrift institutions, given that thrifts compete vigorously with banks for deposits, an important component of liquidity creation. We also include SHARE-L, the local market share of large institutions to allow for the possibility that banks of different sizes may compete differently. To control for local market economic conditions, we include the log of population Ln(POP), the log of population density Ln(DENSITY), and income growth INC-GROWTH.

6. Regression results

In this section, we present our regression results. We first present our main results and find that capital affects liquidity creation differently for large and small banks. We then investigate why the results differ for these two size classes using the components of liquidity creation. Finally, we give the results for large and small banks split by their lagged capital ratios in both size classes. In all cases, we examine whether the findings are consistent with the economic intuition discussed above.

Before proceeding, we note the important distinction between the liquidity creation weight on capital and the regression coefficient on lagged capital. We assign a weight of -0.5 to equity when forming our liquidity creation measures, the dependent variables in the regressions. This does not imply that when we regress the dollar amount of liquidity creation (normalized by GTA) on the lagged equity

¹⁴ In some cases, we use New England County Metropolitan Areas (NECMAs) in place of MSAs, but for convenience, we use the term MSA to cover both MSAs and NECMAs.

¹⁵ We use shares of deposits because this is the only banking service for which geographic location is publicly available.

ratio, EQRAT, the coefficient on EQRAT should necessarily be negative or close to -0.5. Rather, the measured effect depends on bank behavior. For example, if banks with more lagged equity capital extend significantly more illiquid loans and hold significantly fewer liquid assets than banks with lower levels of capital, we may find a positive association between lagged capital and liquidity creation.

6.1. The net effect of capital on liquidity creation for large and small banks

Panels A and B of Table 4 contain the regression results for large banks (GTA exceeding \$1 billion) and small banks (GTA up to \$1 billion), respectively. The first four columns in both panels give the most basic results. The dollar amount of liquidity creation normalized by GTA is regressed on the bank's three-year historical average equity capital ratio, EQRAT, controlling only for bank size as measured by $\ln(\text{GTA})$ and the bank and time fixed effects. The last four columns in both panels show the results of the full specification that includes the remaining control variables. The findings from the partial and full specifications do not differ materially. We therefore concentrate our discussion on the full specification.

The results in Panel A suggest that for large banks, the net effect of capital on liquidity creation is positive and significant when liquidity creation includes off-balance sheet activities, i.e., when we use our “fat” liquidity creation measures (“cat fat” or “mat fat”). The magnitude of the coefficient on the lagged equity capital ratio in the “cat fat” regression, 0.588, suggests that large banks with a 1 percentage point higher equity capital ratio for the prior three years (i.e., an increase in EQRAT of 0.01) create additional liquidity of well over half of a percentage point of a large bank's GTA, which appears to be a substantial effect. Using the “nonfat” measures, capital does not significantly affect liquidity creation, suggesting that off-balance sheet activities constitute an important part of the effect of capital on liquidity creation for large banks. The EQRAT coefficients in the “cat” and “mat” specifications are of similar magnitude, suggesting that use of maturities in place of categories for loans has relatively little impact on the measured net effect of capital.

The results in Table 4, Panel B suggest that small banks with higher capital ratios create less liquidity, in sharp contrast to the positive or insignificant effect of capital found for large banks. All of

the coefficients on the lagged capital ratio are negative and significant at the 1% level, yielding a fairly clear result that is robust across the liquidity creation measures. Using our preferred “cat fat” measure, the magnitude of the coefficient on the lagged equity capital ratio, -0.376, suggests that small banks with a 0.01 higher EQRAT create less liquidity by about a third of a percentage point of their GTA. As for the large banks, the magnitudes of the net effect of capital on liquidity creation are similar for the “cat” and “mat” measures for small banks. However, a key difference for small banks is that the “fat” and “nonfat” magnitudes are also similar. The inclusion of off-balance sheet activities makes little difference to the net effect of capital on liquidity creation, likely reflecting the much smaller role of these activities for small institutions.

In sum, we find that for large banks, capital has a significantly positive effect on liquidity creation when we use measures that include off-balance sheet activities, while this effect is insignificant when we exclude those activities. For small banks, capital has a negative effect on liquidity creation using all of our measures. Thus, the data suggest that the “risk absorption” hypothesis weakly dominates for large banks and the “financial fragility-crowding out” hypothesis strongly dominates for small banks. We next investigate what drives these differences.

6.2. Why is the net effect of capital on liquidity creation different for large and small banks?

To understand more deeply why the effect of capital on liquidity creation differs for large and small banks, we examine how capital affects the individual components of liquidity creation – e.g., liquid, semi-liquid, and illiquid assets. Specifically, we use the individual components based on our “cat fat” liquidity creation measure normalized by GTA as dependent variables in our regressions.

Panels A and B of Table 5 show the coefficients on EQRAT from these regressions for large and small banks, respectively. All of the control variables from the full specification are included in these regressions, but are not shown in the interest of brevity. Importantly, since liquidity creation equals the weighted sum of the individual components (using the $\frac{1}{2}$, 0, and $-\frac{1}{2}$ liquidity creation weights discussed above), the weighted sums of the EQRAT coefficients on the individual liquidity creation

components in Table 5 equal the coefficient on EQRAT using the “cat fat” measure in Table 4.¹⁶ Therefore, the EQRAT coefficients in the individual component regressions help us understand which components of liquidity creation yield the different results for large and small banks.

The results in Table 5 Panel A suggest that for large banks, lagged capital positively influences liquidity creation on the asset side of the balance sheet as well as off the balance sheet. Banks with higher lagged capital ratios have significantly more illiquid assets, fewer liquid assets, and more illiquid guarantees. These positive effects of capital are partially offset by the fact that large banks with higher lagged capital ratios have significantly higher capital ratios in the current period (i.e., the coefficient on EQRAT in the equity/GTA regression is positive and significant).¹⁷ Thus, the positive effect of lagged capital on liquidity creation calculated using our “cat fat” measure in Table 4 Panel A is the net result of the positive effect of lagged capital on assets and illiquid guarantees being larger than the negative effect on current capital. The insignificant effect of lagged capital on liquidity creation calculated using the “cat nonfat” measure in Table 4 Panel A occurs because the positive effects of lagged capital on illiquid guarantees are excluded – the positive effect of lagged capital on assets is approximately offset by the negative effect on current capital.

The findings in Panel A are also consistent with the economic intuition that the “risk absorption” effect is relatively strong for large banks. Higher capital allows large banks to bear significantly more portfolio risk, and the data suggests that they do so. Large banks with higher capital hold more risky illiquid assets such as commercial loans and risky illiquid guarantees such as loan commitments, and fewer relatively safe liquid assets such as treasuries.

Panel B reveals that similar to large banks, small banks with higher lagged capital ratios have significantly higher capital ratios in the current period. However, in stark contrast to large banks, for small banks, capital has a negative effect on liquidity creation on the asset and liability sides of the balance sheet, and essentially no effect on liquidity creation off the balance sheet. Small banks with

¹⁶ For example, for large banks, $\frac{1}{2} \cdot 0.234 + 0 \cdot 0.028 + -\frac{1}{2} \cdot -0.262 + \frac{1}{2} \cdot -0.223 + \dots = 0.588$.

¹⁷ A few other components have statistically significant coefficients, but have no material effect on liquidity creation – the semi-liquid items have 0 weights and the coefficient on derivatives is very small.

higher lagged capital ratios have significantly more liquid assets and fewer liquid liabilities. Thus, the effect of lagged capital on small banks is consistently negative, as opposed to the positive effect for large banks, because the negative effect on current capital is augmented by a negative effect of capital on the asset and liability side, and is not offset by a positive effect off the balance sheet.

The results in Panel B are also consistent with the economic intuition that the “financial fragility-crowding out” effect is relatively strong for small banks. On the asset side, lagged capital does not have a positive effect on illiquid assets, but instead has a positive effect on liquid assets. This is consistent with the spirit of the financial fragility arguments put forth in Diamond and Rajan (2000, 2001). Capital reduces the financial fragility needed to commit to monitoring its borrowers. As a result, banks with higher lagged capital ratios may invest more in liquid assets, rather than increasing their loans. On the liability side, lagged capital has a negative, statistically significant effect on liquid liabilities, consistent with Diamond and Rajan’s (2000, 2001) financial fragility arguments and with the “crowding out” of transactions deposits as in Gorton and Winton (2000).

6.3. The net effect of capital on liquidity creation for large and small banks split by capital

As indicated in the introduction, we expect that for any given bank size, the “risk absorption” effect may be stronger for banks with low capital ratios than for those with higher capital ratios. Low-capital banks have thinner buffers to absorb risks and face greater regulatory, market, and/or owner pressures to control risk taking. As a result, low-capital banks may be forced to reduce risk by holding fewer illiquid assets, fewer illiquid off-balance sheet guarantees, and more liquid assets, activities which also reduce liquidity creation. For these banks, a marginal increase in capital may allow them to have a relatively large increase in liquidity creation as larger buffers and reduced pressures allow them to take on more risk. Thus, we expect capital to have a more positive effect on liquidity creation for banks with low capital ratios in both size classes.

In Table 6, we analyze this by separating large banks into low-, medium-, and high-capital banks based on the bottom, middle, and top thirds of the distribution of EQRAT (mean EQRAT of 0.06, 0.08, and 0.10, respectively) and by following a similar procedure for small banks (mean EQRAT of

0.07, 0.09 and 0.13, respectively). We regress the dollar amount of liquidity creation (calculated using the “cat fat” measure) normalized by GTA on the lagged capital ratio and the full set of control variables for each capital third. Again, only the EQRAT coefficients are shown in the interest of parsimony.

Panel A shows that the net effect of capital is positive and significant only for low-capital large banks. Panel B indicates that the negative effect of capital on liquidity creation is significant for all but low-capital small banks. Both findings are consistent with the suggestion that the “risk absorption” effect – the positive effect of capital on liquidity creation – may be relatively strong for banks of a given size with low capital ratios.

7. Robustness issues

In this section, we address the robustness of our main finding that for large banks, capital has a weak positive effect on liquidity creation based on measures that include off-balance sheet activities and an insignificant effect on liquidity creation based on measures that exclude these activities, and that for small banks, the effect of capital on liquidity creation is strongly negative and significant for all measures. We examine the robustness to four changes: 1) an alternative method to measuring off-balance sheet bank liquidity creation; 2) an alternative way of establishing which assets are securitizable; 3) the use of an alternative capital ratio; and 4) an instrumental variable approach. We show that our main results are qualitatively unchanged.

7.1. Alternative method to measuring off-balance sheet liquidity creation

Our liquidity creation measures are based on the ease, cost, and time for the bank or customer to obtain liquid funds from each bank activity. An alternative would be to use the probability or frequency with which the bank or customers liquefy the obligation and obtain liquid funds. We argue that the ability or option to obtain funds when needed or desired is more important than the actual drawdown frequency. This is also what the theories suggest – banks create liquidity on the balance sheet because they give depositors a liquid claim to their funds (i.e., the option to withdraw funds when needed) instead of forcing them to hold illiquid loans directly (e.g., Diamond and Dybvig 1983). Similarly, banks create

liquidity off the balance sheet through guarantees that allow customers the option to draw down liquid funds when needed (e.g., Kashyap, Rajan, and Stein 2002).

Despite our reservations, we construct a liquidity creation measure that incorporates the frequency with which customers obtain liquid funds on off-balance sheet guarantees. Our alternative liquidity creation measure is identical to our “cat fat” measure, except that we multiply the dollar amount of illiquid off-balance sheet guarantees by 0.30, the observed frequency of drawdown as documented in recent research (Sufi, forthcoming).¹⁸

Using this alternative “cat fat” measure, we find that liquidity creation of the banking sector is 44% lower in 1993 (\$527 billion versus \$947 billion) and 56% lower in 2003 (\$697 billion versus \$1,570 billion) than using our preferred “cat fat” measure (not shown for brevity). The overall pattern of liquidity creation, however, is fairly similar to the “cat fat” pattern.

In Table 7 Panel A we regress the dollar amount of liquidity creation using this alternative “cat fat” measure normalized by GTA on EQRAT and the other exogenous variables. We find that for large banks, the coefficient on EQRAT is positive but not significant, while for small banks, the coefficient on EQRAT is negative and significant. Thus, based on this alternative method to measuring liquidity creation, we obtain broadly consistent results: for large banks, capital has a weak positive effect on liquidity creation, while for small banks, the effect of capital on liquidity creation is strongly negative and significant.

In principle, this methodology could be applied to all bank activities. For example, the drawdown frequency is 1 for loans since customers have already received liquid funds. However, constructing measures using this methodology is difficult, since data on the frequency of drawdown or sale is unavailable for many activities. More importantly, the use of drawdown rates goes directly against the liquidity creation theories, which argue that banks create liquidity by giving customers the option to obtain liquid funds when needed or desired.

¹⁸ Sufi (forthcoming) uses data on letters of credit and loan commitments over 1996-2003, which corresponds closely with our sample period, and finds that conditional on having a letter of credit or a loan commitment, the probability of drawdown over this time period was approximately 30% in every year.

7.2. Alternative way of establishing which assets are securitizable

The amount of liquidity a bank creates is affected by the bank's ability to securitize its assets. Our "cat" liquidity creation measures incorporate this by classifying loan categories that are relatively easy to securitize (residential real estate loans and consumer loans) as semi-liquid and all other loan categories as illiquid. The "mat" measures are based entirely on maturities and hence do not take differences in securitizability into account.

Our "cat" measures do not incorporate, however, the fact that the ability to securitize assets has developed greatly over our sample period. In every loan category, a larger fraction of loans was securitized in 2003 than in 1993. We now construct alternative "cat" liquidity creation measures that take this development into account.

Our alternative "cat" measures are identical to the "cat" measures described in Section 3, except for the way we classify loans. For each loan category, we obtain year-end U.S. Flow of Funds data on the total amount of loans outstanding and the total amount of loans securitized. We use this data to calculate the fraction of loans that has been securitized in the market in each year. Following Loutskina (2006), we then assume that each bank can securitize that fraction of its own loans. To give an example, in 1993, \$3.1 trillion residential and real estate loans were outstanding in the market, and 48.4% of these loans were securitized. If a bank has \$10 million in residential and real estate loans in that year, we assume that 48.4% thereof can be securitized, and hence, we classify \$4.84 million of these loans as semi-liquid and the remainder as illiquid.

We raise two reservations regarding this alternative approach for our purposes. First, it uses the actual amount of securitization, whereas the theories suggest that the ability to securitize matters for liquidity creation, not the amount securitized. Second, this alternative approach assumes that each bank securitizes the same fraction of loans in a particular category, even though in practice major differences may exist across banks. That is, when we assume that 48.4% of all residential and real estate loans can be securitized in 1993, one bank may have securitized virtually its entire residential real estate portfolio in that year, while another bank securitized nothing.

Using this alternative approach, the banking sector created more liquidity, but the growth pattern is similar to the “cat fat” and “cat nonfat” patterns described in Section 4. Based on this alternative “cat fat” measure, liquidity creation equaled \$1.265 trillion in 1993 and increased by about 50% to \$1.896 trillion in 2003. Based on the alternative “cat nonfat” measure, liquidity creation totaled \$0.665 trillion in 1993 and declined slightly to \$0.660 trillion in 2003.

The regression results presented in Table 7 Panel B reinforce our prior findings. That is, for large banks, capital has a weak positive effect on liquidity creation based on measures that include off-balance sheet activities and an insignificant effect on liquidity creation based on measures that exclude these activities. For small banks, the effect of capital on liquidity creation is strongly negative and significant for all measures.

7.3. Alternative capital ratio

In our main analysis, we use EQRAT, the ratio of equity to GTA, as our key exogenous variable. We now replace EQRAT with the ratio of total capital as defined in the Basel I capital standards to GTA. Total capital includes equity plus limited amounts of other financial instruments, such as long-term subordinated debt.^{19,20}

One motivation for using this alternative capital ratio is to see if there is a different effect of regulatory capital from conventional equity capital on liquidity creation. A second motivation is to allow for a broader definition of capital in line with some of the theoretical studies. For example, Diamond and Rajan (2000, 2001) indicate that capital in their analysis may be interpreted as either

¹⁹ Before 1996, banks were not required to report total capital, and from 1996-2000, banks with total assets less than \$1 billion were not required to report total capital if they indicated on the Call Report that their total capital exceeded 8% of adjusted total assets. We estimate the missing numbers using a special Federal Reserve program based on other Call Report information.

²⁰ Note that we do not use the official Basel I total risk-based capital ratio, which is defined as total capital divided by risk-weighted assets, where risk-weighted assets is the weighted sum of assets and off-balance sheet activities, with the weights based on the perceived credit risk of each activity. This capital ratio is clearly endogenous and its use would result in significant bias, since our dependent variable – bank liquidity creation – is also a weighted sum of assets and off-balance sheet activities (as well as liabilities).

equity or long-term debt, sources of funds that cannot run on the bank. The results are not shown in the tables for brevity's sake, but are available upon request.

The results based on the total capital ratio are qualitatively similar to our main results – the net effect of capital on liquidity creation is positive and significant for large banks based on the “cat” measures (insignificant for the “mat” measures), and strongly negative and significant for small banks for all measures.

7.4. Instrumental variable approach

In our main analysis, we use three-year lagged average values of capital to mitigate potential problems of endogeneity. Nevertheless, bank liquidity creation and capital may be jointly determined, and our use of three-year lagged average values may not be sufficient to address this endogeneity problem. We address this issue using an instrumental variable approach.

For instruments, we need variables that are correlated with the amount of lagged capital (once the effects of the other exogenous variables have been netted out), but do not directly affect the amount of liquidity a bank creates. We select four such variables. The first is D-BHC, a dummy variable that equals one if the bank has been part of a bank holding company (BHC) in any of the past three years and zero otherwise. Capital is related to BHC ownership in part because U.S. regulations require holding companies to be a source of strength for the banks they own and also require banks in the same BHC to cross-guarantee each other and provide capital when needed. Holding companies may also inject capital voluntarily, thereby giving the entities in the holding company access to internal capital markets when needed. However, being part of a BHC does not seem likely to directly affect liquidity creation.

Our second instrument is D-DIV, a dummy variable that equals one if the bank paid dividends in any of the past three years and zero otherwise. Regulators will only allow a bank to pay dividends when it is deemed to have sufficient capital and otherwise be in safe and sound condition. Dividend payment directly affects bank capital, but does not seem to be directly related to liquidity creation.

The third instrument is EFF-TAX, the state income tax rate a bank has to pay. Since interest on

debt is tax-deductible while dividend payments are not, banks that operate in states with higher income tax rates are expected to have lower equity ratios, keeping all else equal. Furthermore, there are no reasons to believe that the state income tax rate directly affects liquidity creation. Similarly to Ashcraft (2006), we use the effective income tax rate to be paid on \$1 million in pretax income as our instrument.^{21,22} If a bank operates in multiple states, we use the bank's weighted average state income tax rate, calculated using the share of deposits in each state (relative to the bank's total deposits) as weights.²³

Our fourth instrument is PROV, the three-year lagged average amount of loan and lease loss provisions. Banks are generally required by regulators to create these provisions based on performance problems on loans issued in prior years. These provisions directly reduce capital in the year they occur. Although provisions may be higher at banks with certain loan portfolios (e.g. banks with large C&I portfolios), past provisions should not directly affect the current level of liquidity creation.

Table 8 Panel A shows the results of our first-stage regressions, in which we regress EQRAT on the instruments plus all the exogenous variables for large and small banks. Being part of a BHC has a negative effect on capital, consistent with our hypothesis that banks that are part of a BHC have easier access to capital. It is not surprising that this effect is only significant for small banks, since large banks have easier access to outside capital than small banks, even if they are not part of a BHC. Whether a bank pays dividends does not significantly affect the capital ratio. The state income tax rate has a negative effect on capital for large banks, consistent with Ashcraft (2006). The tax rate does not significantly affect capital at small banks, potentially because the tax benefit of debt may outweighed by

²¹ In each state (except Ohio), the highest tax bracket starts at or below \$1 million in pre-tax profits: when we use the marginal tax rate on \$1 million in pre-tax profits as our instrument we obtain similar results. In Ohio, banks pay 0.015 times the book value of their stock. However, for comparability reasons, we use the corporate income tax rate to calculate Ohio taxes.

²² In contrast to Ashcraft (2006), we use the income tax rate banks have to pay rather than the corporate income tax rate. These rates differ in ten states. To illustrate, in South Dakota, corporations did not pay income tax between 1993 and 2003, while banks paid 6%. In North Dakota, corporations paid 10.35%, while banks were taxed at 7%. Also, unlike Ashcraft (2006), we do not average the tax rate over our sample period. This ensures that we do not use forward-looking data in our regressions.

²³ It would be preferable to use the share of pre-tax income earned in each state as weights, but Call Reports (and other data sources) do not provide this data.

safety and soundness considerations that induce small banks to hold higher capital ratios. Provisions have a positive effect on capital, which is significant for small banks. The positive sign on provisions may occur because banks with riskier loan portfolios hold more capital to absorb risk and/or raise capital around the time of provisioning, so although provisions directly reduce capital, the overall relationship between lagged capital and lagged provisions is positive.

Table 8 Panel B contains the second-stage instrumental variable regression results for large and small banks.²⁴ When we use instruments for capital, our results are consistent with earlier findings. For large banks, the EQRAT coefficients on the “fat” measures remain positive but are not statistically significant, while the EQRAT coefficients on the “nonfat” measures remain insignificant. For small banks, the coefficient on EQRAT is negative and significant in all four specifications.

Note that for both large and small banks, the coefficients on EQRAT are larger when we use instruments, suggesting that the effect of capital on liquidity creation is roughly five times the previously estimated effect. This suggests that in our main liquidity creation regressions, EQRAT is negatively correlated with the residuals, thus inducing a negative bias in our coefficient estimates. When we use instruments for capital, we obtain consistent estimates (see Levitt 1996).²⁵ However, since we had no a priori reason to believe that our EQRAT coefficients were understated in our main regressions, we are hesitant to put too much weight on this explanation.

Thus, our instrumental variable analysis reinforces our main findings. In particular, for large banks, capital has a weak positive effect on liquidity creation based on measures that include off-balance sheet activities and an insignificant effect on liquidity creation based on measures that exclude these activities. For small banks, the effect of capital on liquidity creation is strongly negative and significant for all measures.

²⁴ The results shown use all four instruments. The findings are qualitatively similar if instead we only use the instruments that significantly affect EQRAT (not shown for brevity).

²⁵ Similar logic is used by Levitt (1996), who finds that the number of prisoners has a negative effect on crime that is five times larger when he uses instruments for the prison population.

8. Conclusion

According to banking theory, an important economic role of banks is to create liquidity, but the absence of comprehensive measures of bank liquidity creation makes it difficult to determine the magnitude of liquidity creation and its intertemporal patterns. The first contribution of this paper is the development of four such measures of bank liquidity creation, which we apply to data on U.S. banks over 1993-2003, and analyze how liquidity creation has changed over time. Our second contribution is that we use our liquidity creation measures to study an issue of significant research and policy relevance: the effect of bank capital on liquidity creation.

Our calculations suggest that banks created over \$1.5 trillion in liquidity in 2003 based on our preferred “cat fat” measure. Liquidity creation grew by approximately two-thirds in real terms between 1993 and 2003, despite declining slightly between 2000 and 2003. The dollar amount of liquidity creation is similar using our “mat fat” measure. The liquidity creation pattern over time is also comparable, except that liquidity creation increased again after a dip in 2000. Liquidity creation drops by over 50% when we instead use the two “nonfat” measures. Notably, off-balance sheet activities account for a significant portion of the industry’s liquidity creation.

We examine how capital affects liquidity creation. Some theories predict that higher capital may suppress liquidity creation by reducing financial fragility and/or crowding out liquid deposits (“financial fragility-crowding out” effect). Others suggest that banks with higher capital may create more liquidity because capital gives them greater capacity to absorb the risks associated with liquidity creation (“risk absorption” effect). We find that both sets of theories have empirical validity and that the relationship between banks capital and liquidity creation depends on a bank’s size as well as its capital ratio. As robustness checks we use alternative methods of measuring off-balance sheet liquidity creation, different measures of capital, and an instrumental variable approach that takes the potential endogeneity of capital into account, and find similar results.

Our finding that the effect of bank capital on bank liquidity creation differs across large and small banks raises interesting policy issues. It is well-known that regulators impose capital

requirements on banks for safety and soundness reasons. What our findings suggest is that, while regulators may be able to make banks safer by imposing higher capital requirements, this benefit will have associated with it reduced liquidity creation by small banks, but not by large banks.

Our liquidity creation measures may also be used to address a number of other issues of research and policy relevance that are beyond the scope of this paper, but may be pursued in future research. These include, but are not limited to a comparative analysis of bank liquidity creation across nations, the impact of liquidity creation on economic growth, and an assessment of liquidity created by various financial markets relative to banks.

References

- Allen, Franklin, and Anthony M. Santomero, 1998, The theory of financial intermediation, *Journal of Banking and Finance* 21: 1461-1485.
- Allen, Franklin, and Douglas Gale, 2004, Financial intermediaries and markets, *Econometrica* 72: 1023-1061.
- Ashcraft, Adam B., 2006, Does the market discipline banks? New evidence from the regulatory capital mix, Federal Reserve Bank of New York Staff Reports No. 244.
- Berger, Allen N., and Gregory F. Udell, 1994, Did risk-based capital allocate bank credit and cause a 'credit crunch' in the United States?, *Journal of Money, Credit, and Banking* 26: 585-628.
- Berger, Allen N., and Gregory F. Udell, 2004, The institutional memory hypothesis and the procyclicality of bank lending behavior, *Journal of Financial Intermediation* 13: 458-495.
- Bhattacharya, Sudipto, and Anjan V. Thakor, 1993, Contemporary banking theory, *Journal of Financial Intermediation*, 3: 2-50.
- Calomiris, Charles W., and Charles M. Kahn, 1991, The role of demandable debt in structuring optimal banking arrangements, *American Economic Review* 81: 497-513.
- Deep, Akash, and Guido Schaefer, 2004, Are banks liquidity transformers?, Harvard University working paper (May).
- Diamond, Douglas W., 1984, Financial intermediation and delegated monitoring, *Review of Economic Studies* 51: 393-414.
- Diamond, Douglas W., and Philip H. Dybvig, 1983, Bank runs, deposit insurance, and liquidity, *Journal of Political Economy* 91: 401-419.
- Diamond, Douglas W., and Raghuram G. Rajan, 2000, A theory of bank capital, *Journal of Finance* 55: 2431-2465.
- Diamond, Douglas W., and Raghuram G. Rajan, 2001, Liquidity risk, liquidity creation, and financial fragility: a theory of banking, *Journal of Political Economy* 109: 287-327.
- Flannery, Mark J., 1994, Debt maturity and the deadweight cost of leverage: optimally financing banking firms, *American Economic Review* 84: 320-321.
- Flannery, Mark J., and Kasturi P. Rangan, 2004, What Caused the Bank Capital Build-up of the 1990s?, FDIC Center for Financial Research working paper 2004-03.
- Gambacorta, Leonardo, and Paolo E. Mistrulli, 2004, Does bank capital affect lending behavior?, *Journal of Financial Intermediation* 13: 436-457.
- Gatev, Evan, and Philip E. Strahan, forthcoming, Banks' advantage in hedging liquidity risk: Theory and evidence from the commercial paper market, *Journal of Finance*.
- Gatev, Evan, Til Schuermann, and Philip E. Strahan, 2005, How do banks manage liquidity risk?

- Evidence from the equity and deposit markets in the Fall of 1998, in Mark Carey and Rene Stulz, eds., *Risks of Financial Institutions*, Chicago: University of Chicago Press.
- Gorton, Gary, and Andrew Winton, 2000, Liquidity provision, bank capital, and the macroeconomy, University of Minnesota working paper.
- Hancock, Diana, Andrew J. Laing, and James A. Wilcox, 1995, Bank balance sheet shocks and aggregate shocks: their dynamic effects on bank capital and lending, *Journal of Banking and Finance* 19: 661-77.
- Holod, Dmytro, and Joe Peek, 2004, Asymmetric information and liquidity constraints: a more complete test, University of Kentucky working paper.
- Holmstrom, Bengt, and Jean Tirole, 1998, Public and private supply of liquidity, *Journal of Political Economy* 106:1-40.
- Kashyap, Anil K., Raghuram G. Rajan, and Jeremy C. Stein, 2002, Banks as liquidity providers: an explanation for the coexistence of lending and deposit-taking, *Journal of Finance* 57: 33-73.
- Levitt, Steven D., The effect of prison population size on crime rates: Evidence from prison overcrowding litigation, *Quarterly Journal of Economics* 111: 319-351.
- Loutschina, Elena, 2005, Does securitization affect bank lending? Evidence from bank responses to funding shocks, Boston College working paper.
- Peek, Joe, and Eric S. Rosengren. 1995, The capital crunch: neither a borrower nor a lender be, *Journal of Money, Credit and Banking* 27: 625-38.
- Pennacchi, George, 2006, Deposit insurance, bank regulation, and financial system risks, *Journal of Monetary Economics* 53: 1-30.
- Ramakrishnan, Ram T.S., and Anjan V. Thakor, 1984, Information reliability and a theory of financial intermediation, *Review of Economics Studies* 51: 415-432.
- Repullo, Rafael, 2004, Capital requirements, market power, and risk-taking in banking, *Journal of Financial Intermediation* 13: 156-182.
- Smith, Adam, 1776, *An inquiry into the nature and causes of the wealth of nations*, edited by Edwin Cannan 1976, reproduced at <http://www.econlib.org/library/Smith/smWN.html>.
- Sufi, Amir, forthcoming, Bank lines of credit: an empirical analysis, *Review of Financial Studies*.
- Thakor, Anjan V., 1996, Capital requirements, monetary policy, and aggregate bank lending: theory and empirical evidence, *Journal of Finance* 51: 279-324.
- Von Thadden, Ernst-Ludwig, 2004, Bank capital adequacy regulation under the new Basel Accord, *Journal of Financial Intermediation* 13: 90-95.

Table 1: Liquidity classification of bank activities and construction of four liquidity creation measures

This table explains our methodology to construct liquidity creation measures in three steps.

Step 1: We classify all bank activities as liquid, semi-liquid, or illiquid. For activities other than loans, we combine information on product category and maturity. Due to data limitations, we classify loans entirely by product category (“cat”) or maturity (“mat”).

Step 2: We assign weights to the activities classified in Step 1.

ASSETS:

Illiquid assets (weight = ½)		Semi-liquid assets (weight = 0)		Liquid assets (weight = - ½)
(cat)	(mat)	(cat)	(mat)	
Commercial real estate loans (CRE)	All loans and leases with a remaining maturity > 1 year	Residential real estate loans (RRE)	All loans and leases with a remaining maturity <= 1 year	Cash and due from other institutions
Loans to finance agricultural production		Consumer loans		All securities (regardless of maturity)
Commercial and industrial loans (C&I)		Loans to depository institutions		Trading assets
Other loans and lease financing receivables		Loans to state and local governments		Fed funds sold
	Other real estate owned (OREO)	Loans to foreign governments		
	Customers’ liability on bankers acceptances			
	Investment in unconsolidated subsidiaries			
	Intangible assets			
	Premises			
	Other assets			

LIABILITIES PLUS EQUITY:

Liquid liabilities (weight = ½)	Semi-liquid liabilities (weight = 0)	Illiquid liabilities plus equity (weight = - ½)
Transactions deposits (domestic)	Savings deposits (domestic)	Bank’s liability on bankers acceptances
Overnight federal funds purchased	Time deposits (domestic)	Subordinated debt
Trading liabilities	Foreign deposits	Other liabilities
	Other borrowed money	Equity

OFF-BALANCE SHEET GUARANTEES (notional values):

Illiquid guarantees (weight = ½)	Semi-liquid guarantees (weight = 0)	Liquid guarantees (weight = - ½)
Unused commitments	Net standby letters of credit	Net participations acquired
Commercial and similar letters of credit	Net credit derivatives	
All other off-balance sheet liabilities	Net securities lent	

OFF-BALANCE SHEET DERIVATIVES (gross fair values):

Liquid derivatives (weight = -½)
Interest rate derivatives
Foreign exchange derivatives
Equity and commodity derivatives

Table 1: Liquidity classification of bank activities and construction of four liquidity creation measures – cont'd

Step 3: We combine bank activities as classified in Step 1 and as weighted in Step 2 in different ways to construct four liquidity creation measures by using the “cat” or “mat” classification for loans, and by alternatively including off-balance sheet activities (“fat”) or excluding these activities (“nonfat”).

cat fat =	$+ \frac{1}{2} * \text{illiquid assets (cat)}$ $+ \frac{1}{2} * \text{liquid liabilities}$ $+ \frac{1}{2} * \text{illiquid guarantees}$	$+ 0 * \text{semi-liquid assets (cat)}$ $+ 0 * \text{semi-liquid liabilities}$ $+ 0 * \text{semi-liquid guarantees}$	$- \frac{1}{2} * \text{liquid assets}$ $- \frac{1}{2} * \text{illiquid liabilities}$ $- \frac{1}{2} * \text{equity}$ $- \frac{1}{2} * \text{liquid guarantees}$ $- \frac{1}{2} * \text{liquid derivatives}$
cat nonfat =	$+ \frac{1}{2} * \text{illiquid assets (cat)}$ $+ \frac{1}{2} * \text{liquid liabilities}$	$+ 0 * \text{semi-liquid assets (cat)}$ $+ 0 * \text{semi-liquid liabilities}$	$- \frac{1}{2} * \text{liquid assets}$ $- \frac{1}{2} * \text{illiquid liabilities}$ $- \frac{1}{2} * \text{equity}$
mat fat =	$+ \frac{1}{2} * \text{illiquid assets (mat)}$ $+ \frac{1}{2} * \text{liquid liabilities}$ $+ \frac{1}{2} * \text{illiquid guarantees}$	$+ 0 * \text{semi-liquid assets (mat)}$ $+ 0 * \text{semi-liquid liabilities}$ $+ 0 * \text{semi-liquid guarantees}$	$- \frac{1}{2} * \text{liquid assets}$ $- \frac{1}{2} * \text{illiquid liabilities}$ $- \frac{1}{2} * \text{equity}$ $- \frac{1}{2} * \text{liquid guarantees}$ $- \frac{1}{2} * \text{liquid derivatives}$
mat nonfat =	$+ \frac{1}{2} * \text{illiquid assets (mat)}$ $+ \frac{1}{2} * \text{liquid liabilities}$	$+ 0 * \text{semi-liquid assets (mat)}$ $+ 0 * \text{semi-liquid liabilities}$	$- \frac{1}{2} * \text{liquid assets}$ $- \frac{1}{2} * \text{illiquid liabilities}$ $- \frac{1}{2} * \text{equity}$

Table 2: Summary statistics on bank liquidity creation

Panel A shows liquidity creation of the banking sector, measured using all four liquidity creation measures, in \$ billion and as a fraction of gross total assets (GTA) in 1993 and 2003. Panel B contains graphs of liquidity creation from 1993 – 2003. Both panels show results for all banks, large banks (GTA exceeding \$1 billion), and small banks (GTA up to \$1 billion). All financial values are expressed in real 2003 dollars using the implicit GDP price deflator.

cat (mat): all bank activities other than loans are classified based on product category and maturity; loans are classified by category (maturity) only.
fat (nonfat): off-balance sheet activities are included (excluded).

Panel A: Liquidity creation in 1993 and 2003

Liquidity creation over time (1993 – 2003)

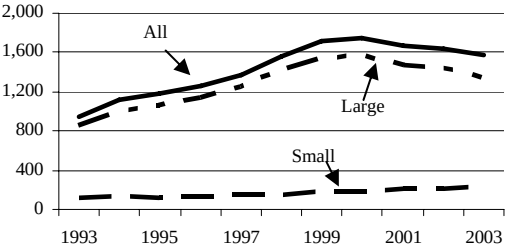
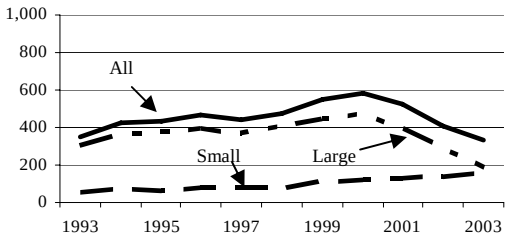
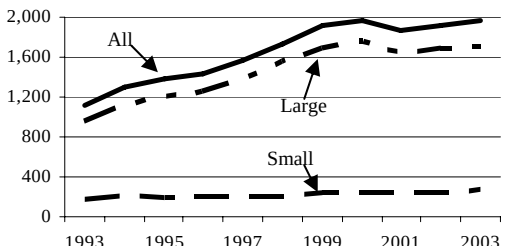
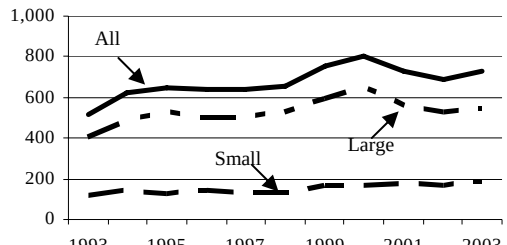
Liquidity creation measure used:		1993 liquidity creation			2003 liquidity creation			Liquidity creation 1993 - 2003	
		N	(\$ billion)	(fraction of GTA)	N	(\$ billion)	(fraction of GTA)	(\$ billion)	
cat fat (preferred)	All banks	9,071	947	0.21	6,959	1,570	0.22		
	Large	411	845	0.26	348	1,340	0.22		
	Small	8,660	103	0.09	6,611	229	0.19		
cat nonfat	All banks	9,071	347	0.08	6,959	334	0.05		
	Large	411	300	0.09	348	183	0.03		
	Small	8,660	47	0.04	6,611	151	0.12		
mat fat	All banks	9,071	1,116	0.25	6,959	1,961	0.27		
	Large	411	944	0.29	348	1,702	0.28		
	Small	8,660	172	0.15	6,611	260	0.21		
mat nonfat	All banks	9,071	516	0.12	6,959	726	0.10		
	Large	411	399	0.12	348	544	0.09		
	Small	8,660	117	0.10	6,611	182	0.15		

Table 3: Definitions and summary statistics for exogenous variables

All exogenous variables are three-year lagged averages (i.e. the average of three years prior to observation of the dependent variable). All of the lagged values are merger-adjusted – the bank capital ratio and size are pro-forma values, the mergers and acquisitions dummies simply take a value of 1 or 0 based on the combined experience of the banks in the case of mergers or acquisitions, and the local market competition and environment variables are weighted averages for the merging banks using their GTA values in constructing the weights. Sample period: 1993 – 2003. All financial values are expressed in real 2003 dollars using the implicit GDP price deflator.

Data sources: Bank Call reports, Bank Holding Company Y-9 reports, FDIC Summary of Deposits, NIC Database, Bureau of Economic Analysis, and U.S. Census Bureau.

Variable	Definition	Mean for all banks	Mean for large banks	Mean for small banks
<u>Bank capital ratio</u>				
EQRAT	Equity capital ratio: total equity capital as a proportion of GTA, where GTA equals total assets plus the allowance for loan and the lease losses and the allocated transfer risk reserve (a reserve for certain foreign loans).	0.10	0.08	0.10
<u>Bank size</u>				
Ln(GTA)	Natural log of GTA.	11.61	15.15	11.44
<u>Mergers and acquisitions</u>				
D-BANK-MERGE	Dummy that equals 1 if the bank was involved in one or more mergers over the past 3 years, combining the charters of two or more banks.	0.09	0.52	0.07
D-DELTA-OWN	Dummy that equals 1 if the bank was acquired in the last 3 years, indicated by a change in top-tier holding company with no change in charter.	0.09	0.08	0.09
<u>Local Market Competition</u>				
HERF	A bank-level Herfindahl index based on bank and thrift deposits (the only variable for which geographic location is publicly available). We first establish the Herfindahl index of the markets in which the bank has deposits and then weight these market indices by the proportion of the bank's deposits in each of these markets.	0.21	0.16	0.21
SHARE-L	Share of market bank and thrift deposits held by large banks (GTA exceeding \$1 billion).	0.32	0.57	0.31
<u>Local market economic environment</u>				
Ln(POP)	Natural log of weighted average population in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights.	11.90	14.03	11.80
Ln(DENSITY)	Weighted average population density (natural log of population per square mile) in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights.	4.69	6.31	4.61
INC-GROWTH	Weighted average income growth in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights.	0.05	0.05	0.05
<u>Fixed effects:</u>				
Time fixed effects	Set of dummies for all but one year.			
Bank fixed effects	Set of dummies for all but one bank.			

Table 4: The effect of capital on liquidity creation for large and small banks

This table presents regression results. The dependent variable is the dollar amount of liquidity a bank has created, calculated using the four liquidity creation measures as defined in Table 1, normalized by GTA. GTA equals total assets plus the allowance for loan and the lease losses and the allocated transfer risk reserve (a reserve for certain foreign loans). Panel A contains the results for large banks (GTA exceeding \$1 billion). Panel B shows the results for small banks (GTA up to \$1 billion).

EQRAT is the equity capital ratio (total equity capital as a proportion of GTA). Ln(GTA) is the log of GTA. D-BANK-MERGE is a dummy that equals 1 if the bank was involved in one or more mergers over the past 3 years, combining the charters of two or more banks. D-DELTA-OWN is a dummy that equals 1 if the bank was acquired in the last 3 years, indicated by a change in top-tier holding company with no change in charter. HERF is a bank-level Herfindahl index based on bank and thrift deposits (the only variable for which geographic location is publicly available). We first establish the Herfindahl index of the markets in which the bank has deposits and then weight these market indices by the proportion of the bank's deposits in each of these markets. SHARE-L is the share of market bank and thrift deposits held by large banks (GTA exceeding \$1 billion). Ln(POP) is the natural log of weighted average population in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights. Ln(DENSITY) is the weighted average population density (natural log of population per square mile) in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights. INC-GROWTH is the weighted average income growth in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights. All regressions are run with both time fixed effects and bank fixed effects (not shown). The sample period is 1993-2003. t-statistics based on robust standard errors clustered by bank are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% level, respectively.

Panel A: Regression results for large banks (GTA exceeding \$1 billion)

	cat fat / GTA	cat nonfat / GTA	mat fat / GTA	mat nonfat / GTA	cat fat / GTA	cat nonfat / GTA	mat fat / GTA	mat nonfat / GTA
EQRAT	0.632 (1.96)**	-0.121 (0.66)	0.686 (2.16)**	-0.067 (0.39)	0.588 (1.78)*	-0.115 (0.63)	0.627 (1.91)*	-0.076 (0.43)
Ln(GTA)	-0.021 (1.89)*	-0.002 (0.47)	-0.010 (0.89)	0.009 (1.71)*	-0.025 (2.17)**	-0.004 (0.86)	-0.014 (1.16)	0.007 (1.32)
D-BANK-MERGE					0.007 (1.31)	0.002 (0.77)	0.010 (1.92)*	0.006 (1.70)*
D-DELTA-OWN					-0.005 (0.26)	-0.017 (2.64)***	0.005 (0.26)	-0.007 (1.08)
HERF					0.106 (0.63)	0.011 (0.13)	0.263 (1.49)	0.168 (2.00)**
SHARE-L					-0.067 (2.63)***	-0.027 (1.79)*	-0.068 (2.50)**	-0.029 (1.69)*
Ln(POP)					0.040 (2.63)***	0.013 (1.72)*	0.029 (1.79)*	0.001 (0.14)
Ln(DENSITY)					-0.051 (3.08)***	-0.023 (2.82)***	-0.036 (2.13)**	-0.008 (0.87)
INC-GROWTH					0.330 (1.55)	-0.140 (1.11)	0.274 (1.24)	-0.196 (1.42)
Constant	0.464 (2.75)***	0.131 (1.81)*	0.359 (2.05)**	0.027 (0.34)	0.277 (1.90)*	0.141 (1.63)	0.221 (1.44)	0.085 (0.87)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3937	3937	3937	3937	3937	3937	3937	3937
Adjusted R-squared	0.71	0.82	0.67	0.80	0.71	0.82	0.67	0.80

Table 4: The effect of capital on liquidity creation for large and small banks – cont'd

Panel B: Regression results for small banks (GTA up to \$1 billion)

	cat fat / GTA	cat nonfat / GTA	mat fat / GTA	mat nonfat / GTA	cat fat / GTA	cat nonfat / GTA	mat fat / GTA	mat nonfat / GTA
EQRAT	-0.354 (10.60)***	-0.365 (12.71)***	-0.415 (14.38)***	-0.425 (17.23)***	-0.376 (11.22)***	-0.385 (13.39)***	-0.439 (15.23)***	-0.448 (18.17)***
Ln(GTA)	-0.003 (1.81)*	-0.005 (3.03)***	-0.022 (12.01)***	-0.024 (14.18)***	-0.009 (4.60)***	-0.010 (5.81)***	-0.028 (14.43)***	-0.029 (16.52)***
D-BANK-MERGE					0.009 (6.58)***	0.009 (7.05)***	0.013 (9.05)***	0.012 (9.59)***
D-DELTA-OWN					0.007 (6.28)***	0.006 (6.35)***	0.005 (4.82)***	0.005 (4.65)***
HERF					0.032 (3.11)***	0.029 (3.06)***	0.009 (0.85)	0.006 (0.57)
SHARE-L					0.014 (4.35)***	0.011 (3.55)***	0.015 (4.39)***	0.012 (3.57)***
Ln(POP)					0.007 (2.39)**	0.007 (2.72)***	0.009 (3.06)***	0.009 (3.30)***
Ln(DENSITY)					0.013 (2.67)***	0.011 (2.53)**	0.006 (1.21)	0.004 (0.90)
INC-GROWTH					0.063 (3.99)***	0.019 (1.36)	0.016 (0.99)	-0.027 (1.88)*
Constant	0.128 (5.80)***	0.114 (5.83)***	0.377 (17.28)***	0.363 (18.39)***	0.043 (1.62)	0.033 (1.36)	0.310 (11.30)***	0.299 (11.91)***
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	80143	80143	80143	80143	80143	80143	80143	80143
Adjusted R-squared	0.85	0.83	0.82	0.80	0.85	0.83	0.82	0.80

Table 5: The effect of capital on the components of liquidity creation

This table presents regression results. The dependent variables are the dollar amounts of the individual liquidity creation components normalized by GTA. The dollar amount of liquidity created is calculated using our “cat fat” liquidity creation measure as defined in Table 1. GTA equals total assets plus the allowance for loan and the lease losses and the allocated transfer risk reserve (a reserve for certain foreign loans).

Panel A contains the results for large banks (GTA exceeding \$1 billion). Panel B shows the results for small banks (GTA up to \$1 billion). Both panels show only the coefficients on EQRAT (total equity capital as a proportion of GTA) in the interest of parsimony, although the regressions include all the exogenous variables from the full specification as defined in Table 3.

The sample period is 1993-2003. All regressions are run with both time fixed effects and bank fixed effects (not shown). t-statistics based on robust standard errors clustered by bank are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% level, respectively.

Panel A: Regression results for large banks (GTA exceeding \$1 billion)

	Assets / GTA			Liabilities / GTA			Equity / GTA	Guarantees / GTA			Derivatives / GTA
	illiquid (cat) $\frac{1}{2}$	semi-liquid (cat) 0	liquid $-\frac{1}{2}$	liquid $\frac{1}{2}$	semi-liquid 0	illiquid $-\frac{1}{2}$	equity $-\frac{1}{2}$	illiquid $\frac{1}{2}$	semi-liquid 0	liquid $-\frac{1}{2}$	liquid $-\frac{1}{2}$
EQRAT	0.234 (1.95)*	0.028 (-0.34)	-0.262 (2.12)**	-0.223 (-1.50)	-0.281 (1.86)*	-0.011 (-0.13)	0.514 (5.50)***	1.398 (2.40)**	-0.332 (1.81)*	0.001 (-0.82)	-0.008 (1.90)*
Observations	3937	3937	3937	3937	3937	3937	3937	3937	3937	3937	3856
Adj. R-squared	0.87	0.84	0.81	0.75	0.74	0.71	0.70	0.51	0.67	0.21	0.22

Panel B: Regression results for small banks (GTA up to \$1 billion)

	Assets / GTA			Liabilities / GTA			Equity / GTA	Guarantees / GTA			Derivatives / GTA
	illiquid (cat) $\frac{1}{2}$	semi-liquid (cat) 0	liquid $-\frac{1}{2}$	liquid $\frac{1}{2}$	semi-liquid 0	illiquid $-\frac{1}{2}$	equity $-\frac{1}{2}$	illiquid $\frac{1}{2}$	semi-liquid 0	liquid $-\frac{1}{2}$	liquid $-\frac{1}{2}$
EQRAT	0.002 (-0.06)	-0.164 (9.73)***	0.163 (6.76)***	-0.259 (18.90)***	-0.091 (5.07)***	-0.011 (2.36)**	0.361 (25.48)***	0.018 (-0.94)	0.001 (-0.30)	0.001 (1.70)*	0.000 (-0.06)
Observations	80143	80143	80143	80143	80143	80143	80143	80143	80143	80143	78669
Adj. R-squared	0.88	0.87	0.81	0.78	0.77	0.68	0.81	0.81	0.50	0.05	0.27

Table 6: The effect of capital on liquidity creation for large and small banks split by lagged capital

This table presents regression results for large and small banks split by capitalization: we separate large banks into low-, medium-, and high-capital banks based on the bottom, middle, and top thirds of the distribution of EQRAT, and follow a similar procedure for small banks. The dependent variable is the dollar amount of liquidity a bank has created normalized by GTA. The dollar amount of liquidity created is calculated using our “cat fat” liquidity creation measure as defined in Table 1. GTA equals total assets plus the allowance for loan and the lease losses and the allocated transfer risk reserve (a reserve for certain foreign loans).

Panel A contains the results for large banks (GTA exceeding \$1 billion). Panel B shows the results for small banks (GTA up to \$1 billion). Both panels show only the coefficients on EQRAT (total equity capital as a proportion of GTA) in the interest of parsimony, although the regressions include all the exogenous variables from the full specification as defined in Table 3.

The sample period is 1993-2003. t-statistics based on robust standard errors clustered by bank are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% level, respectively.

Panel A: Regression results for large banks (GTA exceeding \$1 billion)

	cat fat / GTA			
	All large banks	Low-capital large banks	Medium-capital large banks	High-capital large banks
<i>Mean EQRAT</i>	0.08	0.06	0.08	0.10
EQRAT	0.588 (1.78)*	1.628 (1.96)*	-0.176 (-0.15)	0.558 (-0.72)
Observations	3937	1315	1302	1320
Adjusted R-squared	0.71	0.85	0.84	0.63

Panel B: Regression results for small banks (GTA up to \$1 billion)

	cat fat / GTA			
	All small banks	Low-capital small banks	Medium-capital small banks	High-capital small banks
<i>Mean EQRAT</i>	0.10	0.07	0.09	0.13
EQRAT	-0.376 (11.22)***	0.123 (-1.05)	-0.333 (3.15)***	-0.399 (5.81)***
Observations	80143	26717	26704	26722
Adjusted R-squared	0.85	0.82	0.86	0.89

Table 7: The effect of capital on liquidity creation for large and small banks based on two alternative methods of measuring liquidity creation

This table presents regression results. The dependent variable is the dollar amount of liquidity a bank has created, calculated using the four liquidity creation measures as defined in Table 1, normalized by GTA. GTA equals total assets plus the allowance for loan and the lease losses and the allocated transfer risk reserve (a reserve for certain foreign loans). Panel A shows results for an alternative method to measuring off-balance sheet liquidity creation (discussed in Section 7.1). Panel B shows results for an alternative way to establishing which assets are securitizable (discussed in Section 7.2). Both panels show results for large banks (GTA exceeding \$1 billion) and for small banks (GTA up to \$1 billion).

EQRAT is the equity capital ratio (total equity capital as a proportion of GTA). Ln(GTA) is the log of GTA. D-BANK-MERGE is a dummy that equals 1 if the bank was involved in one or more mergers over the past 3 years, combining the charters of two or more banks. D-DELTA-OWN is a dummy that equals 1 if the bank was acquired in the last 3 years, indicated by a change in top-tier holding company with no change in charter. HERF is a bank-level Herfindahl index based on bank and thrift deposits (the only variable for which geographic location is publicly available). We first establish the Herfindahl index of the markets in which the bank has deposits and then weight these market indices by the proportion of the bank's deposits in each of these markets. SHARE-L is the share of market bank and thrift deposits held by large banks (GTA exceeding \$1 billion). Ln(POP) is the natural log of weighted average population in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights. Ln(DENSITY) is the weighted average population density (natural log of population per square mile) in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights. INC-GROWTH is the weighted average income growth in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights. All regressions are run with both time fixed effects and bank fixed effects (not shown). The sample period is 1993-2003. t-statistics based on robust standard errors clustered by bank are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% level, respectively.

Panel A: Regression results for alternative method to measuring off-balance sheet liquidity creation**Panel B: Regression results for alternative way to establishing which assets are securitizable**

	Large banks	Small banks	Large banks		Small banks	
	cat fat / GTA	cat fat / GTA	cat fat / GTA	cat nonfat / GTA	cat fat / GTA	cat nonfat / GTA
EQRAT	0.098 (0.33)	-0.383 (9.72)***	0.653 (1.58)	-0.050 (0.18)	-0.395 (9.71)***	-0.404 (11.28)***
Ln(GTA)	-0.010 (1.38)	-0.010 (3.16)***	-0.022 (1.63)	0.000 (0.06)	-0.030 (9.42)***	-0.031 (10.88)***
D-BANK-MERGE	0.004 (0.86)	0.009 (4.76)***	0.009 (1.37)	0.004 (1.01)	0.011 (5.81)***	0.011 (6.26)***
D-DELTA-OWN	-0.013 (1.31)	0.006 (4.46)***	0.000 (0.01)	-0.012 (1.34)	0.006 (3.78)***	0.005 (3.69)***
HERF	0.038 (0.30)	0.030 (1.84)*	0.240 (1.32)	0.145 (1.19)	0.035 (2.09)**	0.032 (2.05)**
SHARE-L	-0.039 (1.47)	0.012 (2.21)**	-0.081 (2.13)**	-0.042 (1.56)	0.003 (0.54)	-0.001 (0.13)
Ln(POP)	0.021 (1.67)*	0.007 (1.61)	0.036 (1.43)	0.008 (0.83)	0.008 (1.92)*	0.009 (2.14)**
Ln(DENSITY)	-0.031 (2.26)**	0.011 (1.65)*	-0.045 (1.66)*	-0.017 (1.42)	0.007 (0.95)	0.005 (0.76)
INC-GROWTH	0.000 0.00	0.033 0.036	0.424 (1.24)	-0.046 (0.22)	0.045 (2.09)**	0.002 (0.08)
Constant	3937 0.85	3937 (0.87)	3937 0.77	3937 0.86	3937 (7.28)***	3937 (7.77)***
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bank fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3937	80143	3937	3937	80143	80143
Adjusted R-squared	0.85	0.86	0.77	0.86	0.86	0.85

Table 8: The effect of capital on liquidity creation for large and small banks based on instrumental variable regressions

This table contains results from our instrumental variable approach for large banks (GTA exceeding \$1 billion) and small banks (GTA up to \$1 billion). GTA equals total assets plus the allowance for loan and the lease losses and the allocated transfer risk reserve (a reserve for certain foreign loans). Panel A contains first-stage regression results. The dependent variable is EQRAT, total equity capital as a proportion of GTA. Panel B shows second-stage regression results. In these regressions, the dependent variable is the dollar amount of liquidity a bank has created (calculated using the four liquidity creation measures as defined in Table 1, normalized by GTA) and EQRAT is instrumented with: (1) D-BHC, a dummy that equals 1 if the bank has been part of a bank holding company over the past three years; (2) D-DIV, a dummy that equals 1 if the bank has paid dividends in any of the past three years; (3) EFF-TAX, the effective state income tax rate a bank has to pay on \$1 million in pre-tax income (see Ashcraft, 2006); (4) PROV, the 3-year lagged average amount of provisions.

Ln(GTA) is the log of GTA. D-BANK-MERGE is a dummy that equals 1 if the bank was involved in one or more mergers over the past 3 years, combining the charters of two or more banks. D-DELTA-OWN is a dummy that equals 1 if the bank was acquired in the last 3 years, indicated by a change in top-tier holding company with no change in charter. HERF is a bank-level Herfindahl index based on bank and thrift deposits (the only variable for which geographic location is publicly available). We first establish the Herfindahl index of the markets in which the bank has deposits and then weight these market indices by the proportion of the bank's deposits in each of these markets. SHARE-L is the share of market bank and thrift deposits held by large banks (GTA exceeding \$1 billion). Ln(POP) is the natural log of weighted average population in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights. Ln(DENSITY) is the weighted average population density (natural log of population per square mile) in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights. INC-GROWTH is the weighted average income growth in all markets in which a bank has deposits, using the proportion of deposits held by a bank in each market as weights. All regressions are run with both time fixed effects and bank fixed effects (not shown). The sample period is 1993-2003. t-statistics based on robust standard errors clustered by bank are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% level, respectively.

Panel A: First-stage regression results			Panel B: Second-stage regression results							
	Large banks	Small banks	Large banks				Small banks			
	EQRAT	EQRAT	cat fat	cat nonfat	mat fat	mat nonfat	cat fat	cat nonfat	mat fat	mat nonfat
			/ GTA	/ GTA	/ GTA	/ GTA	/ GTA	/ GTA	/ GTA	/ GTA
EQRAT			2.654	-2.590	4.162	-1.082	-2.109	-1.783	-1.695	-1.369
			(-0.84)	(-0.80)	(-1.07)	(-0.54)	(3.36)***	(3.32)***	(3.34)***	(3.22)***
D-BHC	-0.004	-0.004								
	(-1.00)	(-5.17)***								
D-DIV	0.002	0.000								
	(0.70)	(-0.64)								
EFF-TAX	-0.003	0.000								
	(-1.66)*	(0.02)								
PROV	0.024	0.402								
	(0.46)	(3.58)***								
Ln(GTA)	-0.009	-0.035	-0.007	-0.027	0.017	-0.002	-0.070	-0.060	-0.073	-0.062
	(-3.12)***	(-24.91)***	(-0.25)	(-0.93)	(-0.52)	(-0.11)	(3.19)***	(3.16)***	(4.03)***	(4.08)***
D-BANK-MERGE	0.003	0.007	0.002	0.009	0.001	0.008	0.022	0.019	0.022	0.019
	(3.39)***	(14.96)***	(-0.15)	(-0.93)	(-0.10)	(-1.21)	(4.43)***	(4.47)***	(5.34)***	(5.44)***
D-DELTA-OWN	0.002	0.002	-0.009	-0.013	-0.002	-0.006	0.008	0.007	0.006	0.005
	(0.95)	(4.72)***	(-0.48)	(-1.23)	(-0.08)	(-0.62)	(4.46)***	(4.43)***	(3.59)***	(3.36)***
HERF	0.018	0.010	0.078	0.056	0.211	0.188	0.049	0.042	0.021	0.015
	(0.55)	(2.36)**	(-0.44)	(-0.34)	(-1.01)	(-1.34)	(2.60)***	(2.51)**	(-1.11)	(-0.84)
SHARE-L	-0.005	0.001	-0.060	-0.038	-0.055	-0.033	0.017	0.013	0.017	0.013
	(-0.95)	(0.99)	(-1.62)	(-1.22)	(-1.32)	(-1.11)	(2.80)***	(2.32)**	(2.81)***	(2.28)**
Ln(POP)	0.004	0.002	0.035	0.021	0.018	0.005	0.011	0.010	0.012	0.011
	(1.27)	(1.39)	(-1.44)	(-1.34)	(-0.67)	(-0.31)	(2.00)**	(2.17)**	(2.28)**	(2.35)**
Ln(DENSITY)	-0.002	0.003	-0.047	-0.028	-0.029	-0.010	0.017	0.014	0.009	0.006
	(-0.65)	(1.17)	(1.83)*	(1.93)*	(-1.04)	(-0.61)	(1.95)*	(1.87)*	(-1.07)	(-0.81)
INC-GROWTH	0.027	0.012	0.278	-0.045	0.172	-0.151	0.072	0.027	0.022	-0.022
	(0.78)	(2.24)**	(-0.76)	(-0.18)	(-0.43)	(-0.66)	(2.98)***	(-1.25)	(-0.97)	(-1.06)
Time fixed effects										
Bank fixed effects										
Observations	3823	78888	3823	3823	3823	3823	78888	78888	78888	78888